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# Prediction Interval in Renewable Energy Forecasting: A Comprehensive Review of Uncertainty Quantification Methods

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**ABSTRACT** Robust and reliable forecasting of renewable energy generation is essential for managing grid stability, optimising energy resources, and meeting global climate targets, such as the European Union's goal of achieving Net-Zero greenhouse gas emissions by 2050. Renewable energy sources, including solar, wind, and tidal, are inherently variable and difficult to predict, while hydro is comparatively more stable but still subject to seasonal and climatic fluctuations. This variability poses significant challenges to grid operators and policymakers striving to integrate more renewable energy into the power system. Prediction Interval (PI), which quantifies uncertainty by providing a probabilistic range of potential outcomes, offer a solution to this problem by enhancing the reliability and accuracy of renewable energy forecasts. This paper reviews the methodologies for estimating prediction interval, including statistical, machine learning, and deep learning approaches, and demonstrates how these techniques can address the key challenges in renewable energy forecasting. By offering a structured comparison of prediction interval estimation methods across different renewable energy sectors, the paper highlights their potential to reduce uncertainty, improve decision-making for grid integration, and assist policymakers in developing strategies to mitigate risks from energy variability. Moreover, the review identifies critical gaps in current research and proposes future directions that could further refine Prediction Interval estimation methods for greater accuracy and applicability. The insights provided in this review can aid scientists, engineers, and decision-makers in advancing renewable energy forecasting techniques, thereby accelerating the transition toward a more sustainable and resilient energy system.

**INDEX TERMS** Renewable Energy, Energy Forecasting, Uncertainty Quantification, Prediction Interval, Solar Energy, Wind Energy, Tidal Energy, Hydro Energy

## I. INTRODUCTION

AS the world grapples with the escalating impacts of climate change, the global transition to renewable energy has become a fundamental pillar in the pursuit of sustainable development. The European Union (EU) and other major economies have committed to achieving Net-Zero greenhouse gas (GHG) emissions by 2050, reflecting the urgency of reducing reliance on fossil fuels and shifting to renewable energy sources [1]. The Climate Action Tracker [2] reveals that 89% of total global emissions are now covered by countries that have adopted or announced Net-Zero targets. This includes major emitters such as China, which contributes 28% of global emissions, the United States with 14%, and the European Union with 7%. India and other nations with

similar announcements represent a significant portion of this commitment. However, there remains a notable gap 11% of global emissions are from countries that have yet to commit to Net-Zero targets. This highlights the uneven progress toward climate neutrality, emphasising the need for coordinated global action. Following this, Figure 1 illustrates the share of global GHG emissions covered by countries that have adopted or announced Net-Zero emission targets, emphasising the significant progress made and the gaps that remain.

Renewable energy sources, like solar, wind, tidal, and hydropower are central to achieving these ambitious targets [4]. Their integration into power grids is not merely a climate solution but a technical and logistical necessity for securing a sustainable energy future. However, renewable energy pro-

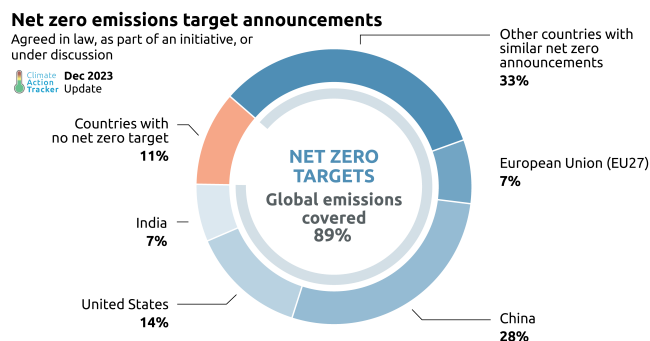


FIGURE 1: Global share of GHG emissions covered by countries with Net-Zero emission targets (source: Climate Action Tracker [2] and IEA Renewables Report [3])

duction is inherently variable, fluctuating based on environmental factors such as weather patterns, geographic location, and seasonal variations [5]. This unpredictability poses a significant challenge for power grid operators, who must balance supply and demand in real time. The variability of renewable energy sources underscores the need for advanced forecasting methods capable of managing this uncertainty, ensuring grid stability, and optimising resource allocation [6].

Accurate forecasting of renewable energy generation is critical to achieving global Net-Zero goals by 2050 [1]. Reliable forecasts are essential for reducing operational risks, optimising energy storage, and ensuring efficient integration of renewable power into the grid. Inaccurate forecasts can result in imbalances that force grid operators to rely on fossil fuel-based backup systems, undermining efforts to reduce carbon emissions [7]. Therefore, the development of advanced forecasting models that account for uncertainty is vital for the effective management of energy systems and the achievement of climate neutrality [8], [9].

The motivation for this research arises from the pressing need to improve renewable energy forecasting in a way that enhances accuracy and effectively quantifies uncertainty. Despite recent advancements in forecasting techniques, significant gaps, particularly in the area of uncertainty quantification [10], [11]. Traditional forecasting models often focus on point predictions, single-value forecasts that fail to capture the range of potential outcomes. This limitation restricts the ability of grid operators and market participants to manage the inherent variability of renewable energy sources.

Prediction Interval (PI) has emerged as a promising solution to this challenge. PI provides a probabilistic range within which future energy generation is expected to fall, accompanied by a specified confidence level [12]. This approach allows for better risk management by giving stakeholders a clearer understanding of possible outcomes and their likelihood. However, the application of PI across different renewable energy sources remains limited and fragmented [13]. The methods for constructing PI often lack the scalability and flexibility required to handle the dynamic and multi-modal

nature of renewable energy data [12], [14], [15]. In particular, the high variability of sources like wind and solar calls for more sophisticated PI estimation techniques that can adapt to changing conditions and capture the complex relationships within the data [16], [17].

## A. RESEARCH OBJECTIVE

This paper seeks to address these gaps by providing a comprehensive review of prediction interval estimation methodologies applied to renewable energy forecasting. The objective is to explore how PI can enhance the reliability and robustness of energy forecasts, helping mitigate the risks associated with energy variability. By systematically examining a range of PI estimation techniques, from traditional statistical methods to advanced machine learning and deep learning models, this paper demonstrates how these techniques can improve grid stability, optimise resource allocation, and support energy market operations. Furthermore, it emphasises the pivotal role of PI estimation in enabling policymakers and energy operators to meet Net-Zero emission targets by 2050.

This research holds significance beyond technical innovation, contributing to the broader goal of accelerating the global transition toward cleaner energy systems. By advancing the methods used for renewable energy forecasting, particularly through the application of Prediction Interval, this work provides critical insights for scientists, policymakers, and industry professionals. PI can enhance forecasting accuracy, improve decision-making processes, and ultimately support the global push toward Net-Zero emissions. As global energy demand rises and the integration of renewable sources into power grids expands, developing more reliable forecasting models will be essential for maintaining grid stability, minimising disruptions, and ensuring a sustainable energy future.

## B. OVERVIEW OF THE PAPER

This paper provides a comprehensive review of prediction interval in renewable energy forecasting. Section II presents an overview of renewable energy sources such as solar, wind, tidal, and hydro, along with the challenges of forecasting each. In Section III, we explore methodologies for quantifying uncertainty in forecasts, including both statistical and machine learning techniques. Section IV focuses on the practical applications of PI and their role in improving the accuracy and reliability of energy forecasts for various renewable sources. Section V discusses the challenges in the field and outlines future research directions, highlighting how PI can address these challenges and drive innovation. Finally, Section VI summarises the key findings and emphasises the crucial role PI plays in the future of renewable energy forecasting.

## C. ABBREVIATIONS AND ACRONYMS

The key abbreviations and acronyms used throughout this paper are summarised in Table 1 for ease of reference.

TABLE 1: Abbreviations and Acronyms

|               |                                                           |              |                                              |
|---------------|-----------------------------------------------------------|--------------|----------------------------------------------|
| <b>AI</b>     | Artificial Intelligence                                   | <b>ANN</b>   | Artificial Neural Network                    |
| <b>BNN</b>    | Bayesian Neural Network                                   | <b>CNN</b>   | Convolutional Neural Network                 |
| <b>DCNN</b>   | Deep Convolutional Neural Network                         | <b>DL</b>    | Deep Learning                                |
| <b>DNN</b>    | Deep Neural Network                                       | <b>ELM</b>   | Extreme Learning Machine                     |
| <b>EMD</b>    | Ensemble Empirical Mode Decomposition                     | <b>GA</b>    | Genetic Algorithm                            |
| <b>GARCH</b>  | Generalised Autoregressive Conditional Heteroscedasticity | <b>GHI</b>   | Global Horizontal Irradiance                 |
| <b>GP</b>     | Gaussian Process                                          | <b>GRU</b>   | Gated Recurrent Unit                         |
| <b>HIRLAM</b> | High Resolution Limited Area Model                        | <b>HIA</b>   | Hybrid Intelligent Algorithm                 |
| <b>HRA</b>    | Harmonic Residual Analysis                                | <b>LSTM</b>  | Long Short-Term Memory                       |
| <b>LUBE</b>   | Lower Upper Bound Estimation                              | <b>ML</b>    | Machine Learning                             |
| <b>MCS</b>    | Monte Carlo Simulation                                    | <b>NN</b>    | Neural Network                               |
| <b>NP</b>     | Neural Process                                            | <b>PI</b>    | Prediction Interval                          |
| <b>PICP</b>   | Prediction Interval Coverage Probability                  | <b>PINAW</b> | Prediction Interval Normalised Average Width |
| <b>PSO</b>    | Particle Swarm Optimisation                               | <b>QDDNN</b> | Quantile-Driven Deep Neural Network          |
| <b>QR</b>     | Quantile Regression                                       | <b>QRF</b>   | Quantile Regression Forest                   |
| <b>RBFNN</b>  | Radial Basis Function Neural Network                      | <b>RNN</b>   | Recurrent Neural Network                     |
| <b>SAEELM</b> | Self-Adaptive Evolutionary Extreme Learning Machine       | <b>SVR</b>   | Support Vector Regression                    |
| <b>TSK</b>    | Takagi-Sugeno-Kang                                        | <b>UQ</b>    | Uncertainty Quantification                   |
| <b>VMD</b>    | Variational Mode Decomposition                            | <b>XAI</b>   | Explainable Artificial Intelligence          |

## II. BACKGROUND ON RENEWABLE ENERGY FORECASTING

### A. INTRODUCTION TO RENEWABLE ENERGY FORECASTING

Renewable energy is at the forefront of the global effort to combat climate change and achieve sustainability. As countries strive to meet ambitious climate goals, such as the European Union's commitment to achieving Net-Zero greenhouse gas emissions by 2050, renewable energy plays a critical role in reducing reliance on fossil fuels [1]. The Climate Action Tracker [2] reveals that 89% of global GHG emissions are now covered by countries that have adopted or announced Net-Zero targets, including major emitters like China, the United States, and the EU. However, significant gaps remain, as 11% of emissions come from countries that have yet to make a commitment to these targets. This emphasises the need for an accelerated transition to renewable energy to achieve global climate neutrality [1], [4].

Renewable energy harnesses naturally replenished resources such as sunlight, wind, water, biomass, and geothermal heat to generate power with minimal environmental impact. These resources, unlike fossil fuels, offer the advantage of generating electricity without emitting harmful greenhouse gases, making them a key solution for mitigating climate change and promoting sustainability [6], [10]. Technologies like solar photovoltaic cells (PV), wind turbines, hydroelectric dams, tidal and wave generators have become essential components of the global renewable energy portfolio, significantly contributing to the reduction of greenhouse gas emissions and the transition to a low-carbon future [12], [18], [19].

Hydropower captures the kinetic energy of flowing water to produce electricity, while biomass energy converts organic matter into heat and power. Geothermal energy utilises the Earth's internal heat for power generation and direct heating, and tidal and wave energy harness the ocean's natural cycles to produce electricity [20], [21]. The growth in renewable energy production over the last two decades has been remark-

able, as illustrated in Figure 2, but the inherent variability of renewable sources like solar and wind presents significant challenges for grid integration. Solar energy generation depends on sunlight, which fluctuates with weather and seasons, while wind energy relies on wind patterns that can be inconsistent. Even more predictable sources like hydropower and tidal energy face seasonal and geographic variability that impacts generation output [19], [22].

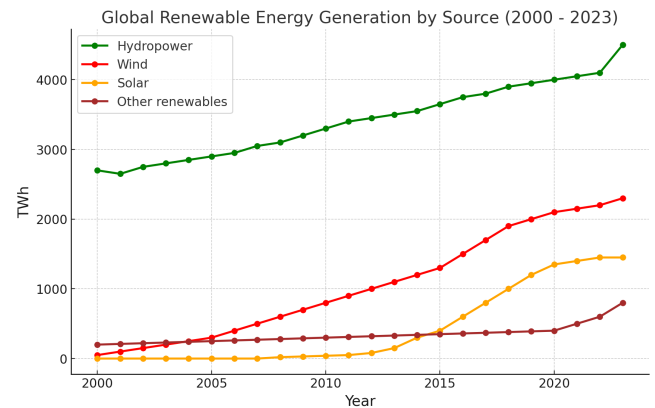


FIGURE 2: Various Renewable Energy Sources - Solar, Wind, Hydro, and Others [3], [23]

Robust forecasting of renewable energy is crucial for maintaining grid stability, optimising energy storage, and ensuring a reliable supply [15], [24]. Forecasting enables grid operators to anticipate fluctuations in energy production, balance supply and demand, and reduce reliance on fossil fuel-based backup systems. Furthermore, accurate forecasts support informed policy decisions, infrastructure investments, and the broader transition to sustainable energy systems capable of meeting global demand while minimising environmental impact [6]. With global climate goals, such as the European Union's Net-Zero emissions target by 2050, renewable energy is poised to play a pivotal role in reducing dependence on fossil fuels [25], [26].

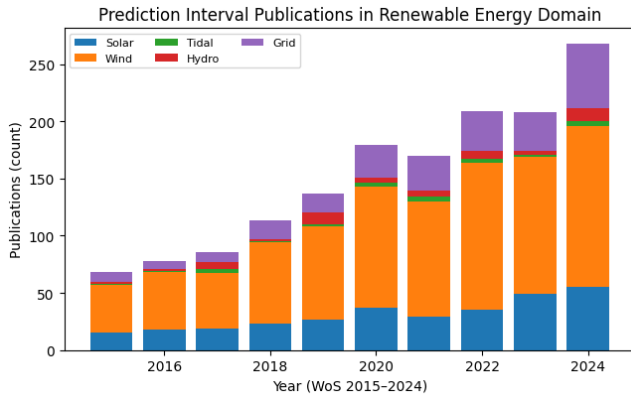


FIGURE 3: Annual publications on Prediction Interval by domain (2015–2024) from Web of Science Core Collection.

To quantify publication trends in prediction-interval and UQ research, we queried the Web of Science Core Collection (Smart Search) and extracted annual counts (final publication year) for solar, wind, tidal, hydro, and grid over 2015–2024; the trend is summarised in Figure 3 [27].

### B. CHALLENGES IN FORECASTING RENEWABLE ENERGY

Forecasting renewable energy generation presents unique challenges due to the inherent variability and intermittency of natural resources. These challenges vary depending on the energy source but generally include environmental, geographic, and technical factors that impact the predictability and reliability of energy output [20], [21], [25].

- 1) **Solar Energy:** Solar energy is highly dependent on weather conditions such as cloud cover, temperature, and solar irradiance, all of which fluctuate throughout the day and across seasons [28]. Geographic location plays a significant role in solar energy generation, as regions closer to the equator receive more consistent sunlight than areas farther from it [20]. The unpredictability of weather patterns introduces significant challenges in forecasting solar energy output. Accurate solar radiation predictions are critical for integrating solar energy into the grid and ensuring a stable electricity supply [17]. The variability of solar irradiance, cloud properties, seasonal changes, and spatial differences introduces uncertainties in forecasting models [5], [29].
- 2) **Wind Energy:** Wind energy also faces challenges related to environmental variability. Wind speed, direction, and air pressure are all influenced by atmospheric conditions and terrain. Wind patterns can change quickly, making short-term forecasts difficult, while long-term forecasts must account for seasonal variations and climate trends [30]. The intermittent nature of wind power can lead to fluctuations in energy generation, impacting grid stability [31], [32]. In addition, wind energy's geographic dependence means that some regions may experience more consistent wind conditions than others, complicating grid integration efforts [33], [34]. Table 2 provides a breakdown of

wind energy capacity by region, highlighting global distribution and trends.

TABLE 2: Wind Power by Country in [3], [35]–[37]

| Country        | Capacity (MW) | Generation (TWh) |
|----------------|---------------|------------------|
| China          | 328,970       | 466.5            |
| United States  | 132,738       | 132.1            |
| Germany        | 63,760        | 341.4            |
| India          | 40,070        | 60.4             |
| Spain          | 27,497        | 56.4             |
| United Kingdom | 27,130        | 75.4             |
| Brazil         | 21,161        | 57.1             |
| France         | 18,676        | 39.7             |
| Canada         | 14,304        | 35.6             |
| Sweden         | 12,080        | 27.5             |

- 3) **Hydropower and Tidal Energy:** Although hydropower and tidal energy are generally more predictable than solar and wind energy, they are still subject to seasonal and geographic influences. Hydropower generation depends on water availability, which can fluctuate based on precipitation, snow melt, and river flow [8], [38]. Tidal energy, while following a consistent cyclical pattern, can be affected by ocean currents and tidal ranges that vary with the phases of the moon and gravitational forces. These factors make it difficult to accurately predict energy generation levels across different time scales [8], [39]. Table 3 presents the various forms of ocean energy sources, along with their capacities and potential generation capabilities, providing insights into the diversity of tidal energy [40], [41]. Table 4 showcases the top tidal power generation companies worldwide, detailing their countries of operation, capacities, and commission years. The importance of forecasting for hydropower is illustrated in Figure 4, which shows the hydropower generation capacities of the top countries worldwide, measured in terawatt-hours (TWh).

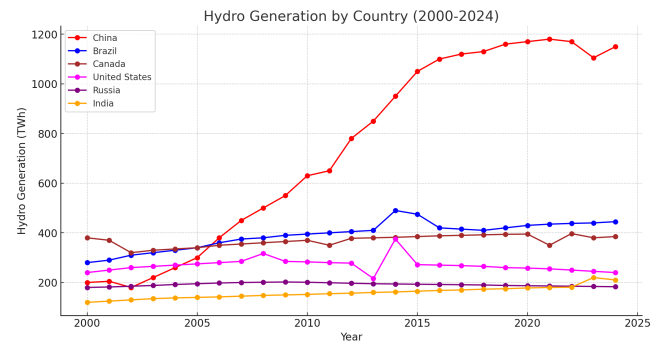


FIGURE 4: Hydropower generation worldwide by top countries, measured in terawatt-hours (TWh) [23] [42].

These challenges significantly impact grid operations, as energy storage and load balancing are required to mitigate the effects of energy variability. Without robust forecasts, grid operators may be forced to rely on backup systems or

TABLE 3: Ocean Energy Sources, Capacity, and Potential Generation [43]–[45]

| Ocean Energy Type               | Capacity (GW) | Potential Generation (TWh) |
|---------------------------------|---------------|----------------------------|
| Tidal                           | 90            | 800                        |
| Marine Current                  | 5,000         | 50,000                     |
| Osmotic Salinity                | 20            | 2,000                      |
| Ocean Thermal Energy Conversion | 1,000         | 10,000                     |
| Sea Wave                        | 1,000–9,000   | 8,000–80,000               |

curtail renewable energy generation, leading to inefficiencies and increased carbon emissions [6], [47]. In addition, the unpredictable nature of renewable energy complicates infrastructure planning and investment decisions, as developers must account for the potential variability in energy supply.

### C. CURRENT FORECASTING METHODS

Forecasting methods for renewable energy generation have evolved significantly over the years, beginning with traditional statistical techniques and advancing to more modern approaches that incorporate machine learning and deep learning [5].

- **Traditional Methods:** Early methods for forecasting renewable energy generation relied on time-series analysis and statistical models, such as Autoregressive Integrated Moving Average (ARIMA), Generalised Autoregressive Conditional Heteroscedasticity (GARCH), and Exponential Smoothing (ETS) [31], [48]. These models are effective for capturing linear relationships in historical data but often struggle with the non-linear and complex dependencies present in renewable energy datasets. Additionally, these models tend to produce point predictions, which do not account for the inherent uncertainty in energy forecasts [13], [20], [49].
- **Machine Learning and Deep Learning Models:** In recent years, advances in Machine Learning (ML) and Deep Learning (DL) have revolutionised renewable energy forecasting [50]. Models such as Artificial Neural Network (ANN), Deep Convolutional Neural Network (DCNN), and Deep Recurrent Neural Network (DRNN) are now widely used for forecasting solar, wind, and hydro energy [20], [51]. These models excel at capturing non-linear relationships and dependencies within complex datasets, making them ideal for renewable energy forecasting, which is influenced by multiple, interrelated factors such as weather patterns and geographic conditions. Machine learning models can also incorporate large volumes of real time data, further improving forecast accuracy [21], [52].

Despite the progress made with machine learning models, point predictions remain limited in their ability to capture uncertainty. Point predictions provide a single estimate of future energy generation but do not convey the range of possible outcomes, leaving grid operators without critical information needed for decision-making under uncertain conditions.

### D. THE ROLE OF PREDICTION INTERVAL IN RENEWABLE ENERGY FORECASTING

Uncertainty is a fundamental challenge in renewable energy forecasting due to the variability of natural resources and environmental conditions. Quantifying uncertainty is crucial for improving the reliability of forecasts and enabling more informed decision-making. prediction interval has emerged as an important tool in addressing this challenge [13], [14].

Prediction Interval provides a range of potential outcomes within a specified confidence level, offering a probabilistic approach to forecasting. Unlike point forecasts, which give a single estimate, PI allows decision-makers to understand the likelihood of different outcomes and manage risk more effectively [53]. For example, a PI for solar energy generation might indicate that with 90% confidence, the output will fall between a certain range, allowing grid operators to plan for various scenarios and allocate resources accordingly [54].

As renewable energy sources take up a larger share of the global energy mix, the need for robust uncertainty quantification becomes more pressing. AI-driven models, such as those based on neural network and fuzzy logic, are increasingly being used to construct PI that can adapt to changing conditions and improve the reliability of forecasts [20], [24], [55]. By incorporating historical data, weather forecasts, and real time monitoring, these models can produce more accurate and reliable predictions, helping energy system operators optimise resource allocation, manage grid stability, and reduce reliance on fossil fuel-based backups [12], [25].

### E. SUMMARY OF THE CHALLENGES OF RENEWABLE ENERGY FORECASTING

Forecasting renewable energy sources continues to face several challenges due to the inherent uncertainties and variability associated with natural resources. These challenges are captured in Tables 5, 6, 7, and 8, which provide a detailed overview of the fundamental issues and benefits across different energy sources, solar [18], wind [33], tidal [56], and hydropower [57]. Key issues include natural variability, forecasting accuracy, grid integration, technological limitations, economic factors, and environmental impacts [49], [58], [59].

Solar energy faces challenges related to the variability of sunlight and meteorological conditions, complicating accurate forecasting and grid integration [60]. Technological advancements are needed to enhance efficiency, reduce costs, and mitigate environmental effects. Similarly, wind energy experiences fluctuations in wind patterns, affecting energy generation and grid stability [28]. Wind power involves high initial investment costs and raises concerns about land use and environmental sustainability [30].

Tidal energy, while more predictable than solar and wind, is hindered by high construction and maintenance costs, as well as potential impacts on marine ecosystems. Scaling tidal infrastructure requires significant financial investment and regulatory support [39]. Hydro energy shares environmental concerns, including habitat disruption and water quality im-

TABLE 4: Top Tidal Power Stations Worldwide [46]

| Station                              | Capacity (MW) | Country       | Commissioned |
|--------------------------------------|---------------|---------------|--------------|
| Sihwa Lake Tidal Power Station       | 254           | South Korea   | 2011         |
| Rance Tidal Power Station            | 240           | France        | 1966         |
| EMEC Fall of Warness Tidal Test Site | 10            | UK (Scotland) | 2007         |
| MeyGen                               | 6             | UK (Scotland) | 2017         |
| Jiangxia Tidal Power Station         | 4.1           | China         | 1980         |
| Kislaya Guba Tidal Power Station     | 1.4           | Russia        | 1968         |

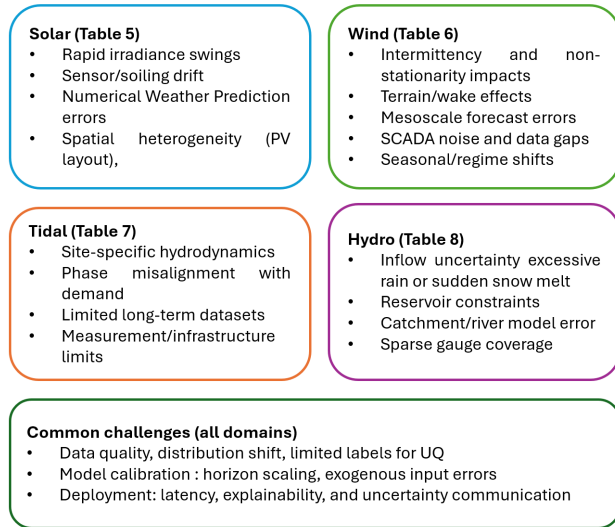


FIGURE 5: Challenge Map summarising key forecasting challenges across renewable domains, distilled from Tables 5–8 (Solar, Wind, Tidal, Hydro).

pacts, in addition to the high infrastructure costs involved in production [38], [41].

Renewable energy sources offer immense potential for reducing carbon emissions and meeting global sustainability goals, the challenges outlined in these tables highlight the critical need for advancements in technology, policy frameworks, and investment strategies [34]. Addressing these issues will improve the accuracy and reliability of renewable energy forecasts, thereby facilitating the successful integration of these sources into global energy systems. For quick reference, we synthesise the domain-specific and common challenges from Tables 5, 6, 7, 8 into a single infographic in Figure 5.

### III. UNCERTAINTY QUANTIFICATION IN RENEWABLE ENERGY FORECASTING

Uncertainty quantification (UQ) is a critical aspect of renewable energy forecasting, enhancing the accuracy and reliability of predictions. In fields like Artificial Intelligence (AI), machine learning (ML), and Deep Learning (DL), UQ provides insights into the confidence and limitations of forecasts by accounting for potential variability and errors in outcomes [13], [72]. In the context of renewable energy, where generation is influenced by fluctuating environmental conditions, UQ is essential for making informed decisions, managing

risks, and optimising resource allocation [55].

Prediction Interval (PI) is a primary method used for UQ in renewable energy forecasting. PI quantifies the uncertainty surrounding energy production estimates by offering a range of possible outcomes with a given confidence level [15]. Unlike point forecast, which gives a single prediction, PI accounts for variables such as weather fluctuations, system performance, and geographic variability, offering a more comprehensive understanding of potential future energy generation [14]. This is crucial for effective risk management, decision-making, and resource planning, particularly in solar and wind energy forecasting [15], [55].

#### A. PREDICTION INTERVAL AND UNCERTAINTY QUANTIFICATION ACROSS RENEWABLE ENERGY SOURCES

Prediction interval is not itself a forecasting algorithm but rather a framework for quantifying uncertainty. Methods such as Quantile Regression (QR) and Lower Upper Bound Estimation (LUBE) are specifically designed to compute PI. Other families of methods, including Bayesian Neural Network (BNN), ensemble learning, and deep learning variants, are model classes that can be adapted to generate PI. It is important to distinguish between PI as a statistical construct and the various methods employed to estimate it, as summarised in Table 9.

PI is an indispensable tool in renewable energy forecasting, especially for sources like solar, wind, tidal, and hydro energy, where variability and uncertainty are significant. By providing upper and lower bounds for future energy outputs, PI offers a probabilistic framework for decision-making [55], [73]. Several probabilistic generation models and uncertainty modelling approaches have also been proposed for solar, wind, and PV-based distributed generation [61], [74]–[77]. In solar energy forecasting, PI accounts for uncertainties such as solar irradiance variability, cloud cover, and atmospheric conditions, which are difficult to predict using single-point estimates. These uncertainties are critical to grid stability and resource optimisation [54], [78].

Similarly, in wind energy forecasting, PI captures fluctuations in wind speed and direction caused by atmospheric changes [30]. Wind energy's inherent variability necessitates considering a range of possible outcomes, making PI an essential method for improving forecast reliability and supporting wind energy integration into the grid [6], [79].

For tidal and hydro energy, while the sources are generally more predictable due to the cyclical nature of tides and consistent water flow, seasonal changes, geographic conditions, and

TABLE 5: Challenges and Advantages of Solar Energy

| Energy Source | Description                                                                                                                                                                                                                                         | Key Advantages                                                                                                                                                                         | Major Challenges                                                                                                                                                                                | References    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Solar         | <ul style="list-style-type: none"> <li>- Variability of solar irradiance and meteorological factors introduces uncertainty into forecasting.</li> <li>- Need for timely and efficient forecasting poses computational challenges.</li> </ul>        | <ul style="list-style-type: none"> <li>- Renewable and abundant resource.</li> <li>- Low environmental impact.</li> <li>- Suitable for grid management and energy planning.</li> </ul> | <ul style="list-style-type: none"> <li>- Variability of solar resources.</li> <li>- Weather dependency.</li> <li>- Forecast horizon limitations.</li> <li>- Data quality issues.</li> </ul>     | [7] [60] [61] |
| Solar         | <ul style="list-style-type: none"> <li>- Optimisation of device efficiencies crucial for sustainable conversion technologies.</li> <li>- Challenges include improving performance, durability, and cost-effectiveness.</li> </ul>                   | <ul style="list-style-type: none"> <li>- Maximises energy harvested from sunlight.</li> <li>- Supports transition to sustainable energy.</li> </ul>                                    | <ul style="list-style-type: none"> <li>- Inefficient utilisation of solar spectrum.</li> <li>- Intermittent and diffuse sunlight.</li> <li>- Need for cost-effective photocatalysts.</li> </ul> | [5] [62]      |
| Solar         | <ul style="list-style-type: none"> <li>- Optimisation techniques crucial for efficiency, cost-effectiveness, and performance.</li> <li>- Enables integration into existing grid infrastructure.</li> <li>- Reduces environmental impact.</li> </ul> | <ul style="list-style-type: none"> <li>- Higher energy output.</li> <li>- Minimised operational costs.</li> <li>- Improved reliability and resilience.</li> </ul>                      | <ul style="list-style-type: none"> <li>- Environmental impact of panel disposal.</li> <li>- Carbon footprint.</li> <li>- Scarcity of training in developing countries.</li> </ul>               | [29]          |
| Solar         | <ul style="list-style-type: none"> <li>- Future directions include advanced forecasting models, real time weather data integration, energy storage optimisation, and microgrid forecasting.</li> </ul>                                              | <ul style="list-style-type: none"> <li>- Enhanced efficiency, reliability, and sustainability of energy systems.</li> </ul>                                                            | <ul style="list-style-type: none"> <li>- Capturing and quantifying uncertainty in prediction interval.</li> <li>- Model generalisation across diverse settings.</li> </ul>                      | [17] [63]     |

TABLE 6: Challenges and Advantages of Wind Energy

| Energy Source | Description                                                                                                                                                                                                                                                             | Key Advantages                                                                                                                                                                  | Major Challenges                                                                                                                                                                                                    | References          |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Wind          | <ul style="list-style-type: none"> <li>- Impact of extreme wind speeds, icing on turbines, sea ice, and permafrost poses challenges.</li> </ul>                                                                                                                         | <ul style="list-style-type: none"> <li>- Abundant and renewable resource.</li> <li>- Potential for increased wind speed extremes.</li> </ul>                                    | <ul style="list-style-type: none"> <li>- Intermittency of wind power generation.</li> <li>- Fluctuations in power output.</li> <li>- Reactive power support.</li> <li>- Voltage and frequency stability.</li> </ul> | [30] [33] [47] [64] |
| Wind          | <ul style="list-style-type: none"> <li>- Understanding and mitigating these challenges are crucial for reliability and efficiency.</li> </ul>                                                                                                                           | <ul style="list-style-type: none"> <li>- Reliability and efficiency with proper mitigation.</li> </ul>                                                                          | <ul style="list-style-type: none"> <li>- Power quality issues.</li> <li>- Fault ride-through capability.</li> <li>- Electricity market dynamics.</li> <li>- Environment considerations.</li> </ul>                  | [64] [65] [66]      |
| Wind          | <ul style="list-style-type: none"> <li>- Consideration of environmental factors crucial for wind farm development.</li> <li>- Understanding and addressing environmental concerns essential for sustainability.</li> </ul>                                              | <ul style="list-style-type: none"> <li>- Reduction of adverse impacts on ecosystems and wildlife.</li> <li>- Promotion of sustainable wind energy projects.</li> </ul>          | <ul style="list-style-type: none"> <li>- Bird and bat mortality.</li> <li>- Noise disturbance.</li> <li>- Visual impact on landscapes.</li> <li>- Effects on climate and communication services.</li> </ul>         | [64] [65]           |
| Wind          | <ul style="list-style-type: none"> <li>- International market expansion driven by environmental awareness and technological advancements.</li> <li>- Wind power contributes to reducing greenhouse gas emissions and enhancing energy security.</li> </ul>              | <ul style="list-style-type: none"> <li>- Diversification of energy mix.</li> <li>- Reduction of greenhouse gas emissions.</li> <li>- Enhancement of energy security.</li> </ul> | <ul style="list-style-type: none"> <li>- Uncertainty in wind energy forecasting.</li> <li>- Variability management and grid integration.</li> </ul>                                                                 | [33]                |
| Wind          | <ul style="list-style-type: none"> <li>- Challenges include intermittency, effective energy storage, environmental impacts, and grid integration constraints.</li> </ul>                                                                                                | <ul style="list-style-type: none"> <li>- Abundant renewable resource.</li> <li>- Reduction of environmental impact.</li> <li>- Enhanced energy security.</li> </ul>             | <ul style="list-style-type: none"> <li>- Intermittency of wind resources.</li> <li>- Grid integration constraints.</li> <li>- Environmental impacts.</li> </ul>                                                     | [67] [68]           |
| Wind          | <ul style="list-style-type: none"> <li>- Non-stationary, nonlinear, and non-normal characteristics of wind speed data pose challenges.</li> <li>- Accurate wind speed prediction crucial for addressing grid frequency maintenance and power quality issues.</li> </ul> | <ul style="list-style-type: none"> <li>- Improved power system planning and operations.</li> <li>- Enhanced grid stability and reliability.</li> </ul>                          | <ul style="list-style-type: none"> <li>- Non-stationarity, nonlinearity, and non-normality of wind speed data.</li> <li>- Accuracy of wind speed prediction.</li> </ul>                                             | [30]                |

TABLE 7: Challenges and Advantages of Tidal Energy

| Energy Source | Description                                                                                                                                                                                                                                                                                                | Key Advantages                                                                                                                          | Major Challenges                                                                                                                                         | References                   |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Tidal         | <ul style="list-style-type: none"> <li>- Challenges primarily revolve around environmental impacts and high initial capital costs.</li> <li>- Deployment can disrupt marine ecosystems and wildlife habitats.</li> <li>- Limited suitable locations due to estuaries and shallow-water straits.</li> </ul> | <ul style="list-style-type: none"> <li>- Abundant and renewable resource.</li> <li>- Reliable and predictable energy source.</li> </ul> | <ul style="list-style-type: none"> <li>- Environmental impacts.</li> <li>- High initial capital costs.</li> <li>- Limited suitable locations.</li> </ul> | [8] [19] [38] [39] [40] [41] |

TABLE 8: Challenges and Advantages of Hydro Energy

| Energy Source | Description                                                                                                                                                                                                                                                                                                                                                                                                                           | Key Advantages                                                                                                                                      | Major Challenges                                                                                                                                                                                                                                                                                                       | References                         |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Hydro         | <ul style="list-style-type: none"> <li>- Various challenges including environmental impacts from dam construction, climate change affecting water availability, operational complexities, uncertainties in flow patterns, and high initial investment costs.</li> <li>- Balancing energy generation with environmental considerations, meeting regulatory standards, and addressing economic viability are key challenges.</li> </ul> | <ul style="list-style-type: none"> <li>- Reliable and renewable energy source.</li> <li>- Long lifespan of hydroelectric infrastructure.</li> </ul> | <ul style="list-style-type: none"> <li>- Environmental impacts from dam construction.</li> <li>- Climate change affecting water availability.</li> <li>- Operational complexities.</li> <li>- Limited equipment capabilities.</li> <li>- Regulatory constraints.</li> <li>- Uncertainties in flow patterns.</li> </ul> | [10] [49] [57] [58] [69] [70] [71] |

unexpected environmental factors still introduce uncertainty [40], [80]. PI offers a robust method for accounting for these variations, ensuring that energy production estimates are reliable even under fluctuating conditions [81].

While Confidence Interval (CI) describe the uncertainty around the estimate of a model's mean [82], prediction interval quantify the range within which future individual observations are expected to fall [55]. In renewable energy forecasting, PI is more appropriate because it directly reflects operational uncertainty faced by grid operators [83]. Thus, PI captures both aleatoric (data) and epistemic (model) uncertainty, whereas CI only addresses estimation of expected values.

## B. METHODS FOR UNCERTAINTY QUANTIFICATION

Several advanced methods have been developed to address uncertainties in renewable energy forecasting. These methods range from traditional statistical techniques to modern machine learning and optimisation approaches, each offering unique advantages in handling different types of uncertainty: aleatoric (inherent variability) and epistemic (model uncertainty) [12], [13].

Uncertainty quantification plays a vital role in enhancing the reliability and robustness of renewable energy forecasts [13], [84]. Several advanced methods have been developed to address the inherent uncertainties in energy generation from variable sources like solar, wind, tidal, and hydropower [68]. These methods range from traditional statistical techniques to modern machine learning and optimisation approaches, each offering unique advantages in handling different types of uncertainty, such as aleatoric and epistemic. Below is a detailed discussion of the key UQ methods used in renewable energy forecasting.

- **Prediction Interval (PI):** PI provides a range within which future observations are expected to fall, offering a probabilistic approach to uncertainty quantification. PI captures both model and data uncertainty, making them a valuable tool for decision-making in renewable energy systems. By providing upper and lower bounds for energy forecasts, PI allows grid operators and policymakers to anticipate the range of possible outcomes, mitigating the risks associated with over or under prediction [12], [53].
- **Lower and Upper Bound Estimation (LUBE):**  
The LUBE method is widely applied in solar and

wind energy forecasting to construct PI. It uses optimisation techniques, such as Extreme Learning Machines (ELM) and bi-level optimisation, to estimate the lower and upper bounds of predictions. This method has been particularly effective in improving the reliability of energy forecasts by accounting for variability in production [12], [14].

- **Quantile Regression (QR):** QR is a non-parametric method that constructs PI based on the quantiles of the forecast distribution, making it especially useful for renewable energy sources like wind and solar, where output distributions are often non-standard. QR helps capture the variability in energy generation without assuming a specific distribution [85]. Recent work has advanced quantile regression with deep neural architectures, including evolutionary non-crossing multi-output designs for ultra-short-term wind forecasting [86].

- **Monte Carlo Simulation (MCS):** MCS is a robust method for UQ that involves randomly sampling input parameters to propagate uncertainties through a model. By generating a statistical distribution of potential outcomes, MCS provides a probabilistic understanding of future energy generation. It has been widely used in wind and hydropower forecasting, where complex and dynamic environmental conditions lead to high uncertainty [13], [87].
- **Bayesian Methods:** Bayesian approaches are particularly useful for UQ in renewable energy forecasting, as they provide a probabilistic framework for updating predictions based on prior knowledge and observed data. Bayesian Neural Network (BNN) extend traditional neural network by treating model parameters as probability distributions, allowing for a better representation of uncertainty in the predictions [88].
- **Bayesian Neural Network (BNN):** BNN treats model parameters as distributions rather than fixed values, allowing for a better representation of uncertainty in renewable energy forecasts. BNN has been applied in solar and wind energy forecasting, accounting for weather-related variability [88].
- **Bayesian Deep Learning:** Bayesian Deep Learning approaches, such as variational inference and Monte Carlo dropout, provide both point estimates

and uncertainty bounds, making them particularly useful in complex systems like wind energy forecasting [89]. Beyond Bayesian methods, conformal prediction offers a distribution-free framework for interval estimation, providing finite-sample guarantees under mild assumptions [90].

- **Ensemble Method:** Ensemble method improves forecast accuracy by combining predictions from multiple models. Bagging and Boosting techniques create multiple versions of a predictor and combine them to form a more accurate prediction. Quantile Regression Forest (QRF) is an extension of ensemble methods specifically designed for UQ, providing a non-parametric approach to constructing PI for renewable energy forecasts [85], [91].
  - **Bagging and Boosting:** These techniques improve predictions by averaging multiple models (bagging) or iteratively refining weak models (boosting). They are widely used in solar and wind energy forecasting to enhance the reliability of PI [91].
  - **Quantile Regression Forest (QRF):** QRF is an ensemble method that predicts multiple quantiles of the forecast distribution, providing a non-parametric approach to constructing PI for solar and wind energy [85].
- **Gaussian Process (GP):** GP provides a distribution over possible functions that fit the data, offering both point estimates and uncertainty bounds. In renewable energy forecasting, GP has been used extensively in wind and solar energy to offer insights into the confidence of predictions and the variability of inputs like wind speed and solar irradiance [92].
- **Neural Network and Deep Learning:** Deep learning models, such as Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) network, are capable of capturing complex patterns in renewable energy data. When combined with UQ techniques, these models provide more accurate predictions by accounting for uncertainty in both the data and the models [20], [93].
  - **Neural Process (NP):** NP incorporates uncertainty through latent variables, allowing for a flexible approach to modelling stochastic processes. NP has been applied to wind energy forecasting, providing probabilistic forecasts that capture the variability in energy generation [94].
  - **Deep Convolutional Neural Network (DCNN):** DCNN is effective for capturing spatial and temporal patterns in renewable energy data. It has been applied in solar and wind energy forecasting, improving the accuracy of PI by modelling the non-linear dependencies between variables [20], [93].
- **Surrogate Model:** Surrogate model provides an efficient way to approximate complex simulation models, reducing computational costs while maintaining accuracy. This model is particularly useful for large-scale

simulations in renewable energy systems, such as wind turbine fatigue loads and energy production forecasts [87].

We present a concise overview of the key uncertainty quantification methods employed in renewable energy forecasting. Table 9 summarises the advantages, limitations, and specific applications of these methods across various renewable energy sources, showcasing their significance in enhancing forecasting accuracy and reliability.

### C. EVOLUTION OF PREDICTION INTERVAL AND UQ IN RENEWABLE ENERGY

Over time, UQ methods have evolved with advancements in Machine Learning (ML) and Deep Learning (DL) models, particularly in the context of renewable energy forecasting [13]. Earlier approaches, such as ARIMA and GARCH [101], were effective but struggled to account for the non-linear and highly variable behaviors of renewable energy systems. Machine learning techniques, such as Bayesian inference [24], Monte Carlo (MC) simulation [29], [100], and Extreme Learning Machine (ELM) [78], have significantly improved the accuracy and robustness of prediction interval in renewable energy forecasting. By incorporating historical data and environmental factors, these models generate more reliable forecasts, enhancing grid stability and supporting effective energy management [67].

For solar energy, ML models like ELM and Quantile Regression [102] have been particularly useful in refining PI. These models utilise historical solar data and meteorological forecasts to produce more accurate and reliable prediction interval, accounting for fluctuations in solar irradiance and other environmental variables [26]. Similarly, wind energy forecasting has benefited from advanced methods such as Lower and Upper Bound Estimation (LUBE) [78] and Bayesian Neural Network (BNN), which generate robust PI that effectively handles the variability in wind speeds [63].

In hydro and tidal energy systems, Monte Carlo simulation and Bayesian method have proven effective in estimating water flow and tidal movements, factoring in seasonal variations and geographic conditions. These approaches improve the management of uncertainties in energy production, making them highly valuable for these renewable energy sources [51], [87].

The successful implementation of UQ methods like PI across various renewable energy systems has significantly enhanced the accuracy and reliability of energy forecasts. Prediction interval is the uncertainty output; methods such as quantile regression, LUBE, Bayesian models, and neural architectures (e.g., ELM/SAEELM/LSTM) are employed to construct it. For instance, in wind power generation, PI improves forecast precision by offering a range of potential outputs, allowing for better risk assessment and decision-making [103]. Techniques like Extreme Learning Machines (ELM) and Self-Adaptive Evolutionary ELM (SAEELM) have been applied to generate reliable PI for wind energy, further supporting the integration of renewable energy into

TABLE 9: Summary of Uncertainty Quantification Methods in Renewable Energy Forecasting

| Method                                                                 | Type             | Advantages                                                                                                                                                                                                                                                                            | Challenges                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conventional Feed-Forward Neural Network</b><br>[12] [14] [15] [13] | Machine Learning | <ul style="list-style-type: none"> <li>- Provides bounds for energy forecasts.</li> <li>- Useful for decision-making and risk management.</li> </ul>                                                                                                                                  | <ul style="list-style-type: none"> <li>- Requires highly accurate underlying model.</li> <li>- Computational complexity increases with data size.</li> </ul>                                                                                                              |
| <b>Lower and Upper Bound Estimation (LUBE)</b><br>[15] [12] [95]       | Statistical      | <ul style="list-style-type: none"> <li>- Narrow prediction interval with high coverage probability.</li> <li>- Does not rely on distribution assumptions.</li> </ul>                                                                                                                  | <ul style="list-style-type: none"> <li>- Sensitive to input data quality.</li> <li>- Requires tuning of custom loss functions.</li> </ul>                                                                                                                                 |
| <b>Quantile Regression (QR)</b><br>[26] [29] [86]                      | Statistical      | <ul style="list-style-type: none"> <li>- Non-parametric method capturing different levels of uncertainty.</li> <li>- Robust against outliers.</li> </ul>                                                                                                                              | <ul style="list-style-type: none"> <li>- Requires large datasets.</li> <li>- Computationally intensive for multiple quantile estimates.</li> </ul>                                                                                                                        |
| <b>Regression Trees</b><br>[96]                                        | Statistical      | <ul style="list-style-type: none"> <li>- Utilises decision trees as predictive models, with leaves representing target variable values.</li> <li>- Generates probabilistic forecasts and prediction interval.</li> </ul>                                                              | <ul style="list-style-type: none"> <li>- Suitable for handling large datasets with high dimensionality.</li> <li>- May suffer from over-fitting, especially with noisy data.</li> </ul>                                                                                   |
| <b>Monte Carlo Simulation (MCS)</b><br>[87]                            | Probabilistic    | <ul style="list-style-type: none"> <li>- Provides a distribution of possible outcomes.</li> <li>- Widely applicable.</li> </ul>                                                                                                                                                       | <ul style="list-style-type: none"> <li>- Computationally expensive for large datasets.</li> <li>- Requires many simulation runs for accuracy.</li> </ul>                                                                                                                  |
| <b>Bayesian Neural Network (BNN)</b><br>[88]                           | Bayesian         | <ul style="list-style-type: none"> <li>- Accounts for model uncertainty.</li> <li>- Flexible in updating predictions with new data.</li> </ul>                                                                                                                                        | <ul style="list-style-type: none"> <li>- Requires significant computational power.</li> <li>- Can be sensitive to prior assumptions.</li> </ul>                                                                                                                           |
| <b>Ensemble Method</b><br>[91]                                         | Machine Learning | <ul style="list-style-type: none"> <li>- Improves robustness by combining multiple models.</li> <li>- Captures both model and data uncertainty.</li> </ul>                                                                                                                            | <ul style="list-style-type: none"> <li>- Computationally expensive, requiring multiple models.</li> <li>- High resource consumption.</li> </ul>                                                                                                                           |
| <b>Gaussian Process (GP)</b><br>[92]                                   | Probabilistic    | <ul style="list-style-type: none"> <li>- Provides distribution over functions with uncertainty bounds.</li> <li>- Models non-linear relationships well.</li> </ul>                                                                                                                    | <ul style="list-style-type: none"> <li>- High computational cost for large datasets.</li> <li>- Struggles with high-dimensional data.</li> </ul>                                                                                                                          |
| <b>Deep Learning (LSTM, CNN, RNN)</b><br>[20]                          | Deep Learning    | <ul style="list-style-type: none"> <li>- Captures complex non-linear dependencies.</li> <li>- Suitable for time series data.</li> </ul>                                                                                                                                               | <ul style="list-style-type: none"> <li>- Prone to over fitting and requires large datasets.</li> <li>- Computationally intensive and requires tuning.</li> </ul>                                                                                                          |
| <b>Bootstrap Method</b><br>[97] [63]                                   | Machine Learning | <ul style="list-style-type: none"> <li>- Estimates aleatoric and epistemic uncertainty separately.</li> <li>- Provides robust uncertainty estimates.</li> </ul>                                                                                                                       | <ul style="list-style-type: none"> <li>- Only outputs symmetric interval.</li> <li>- Requires multiple model trainings, increasing complexity.</li> </ul>                                                                                                                 |
| <b>Artificial Neural Network (ANN)</b><br>[52] [98] [99]               | Machine Learning | <ul style="list-style-type: none"> <li>- Capture complex non-linear relationships in the data.</li> <li>- Adapt to changing patterns and trends in the data.</li> <li>- Process multiple inputs simultaneously.</li> </ul>                                                            | <ul style="list-style-type: none"> <li>- The inner workings models can be complex and difficult to interpret.</li> <li>- Models are prone to over fitting, especially with large and complex datasets.</li> <li>- Require a large amount of data for training.</li> </ul> |
| <b>Monte Carlo Dropout</b><br>[13] [29] [100]                          | Machine Learning | <ul style="list-style-type: none"> <li>- Provides a computationally efficient way to estimate model uncertainty by leveraging dropout during inference.</li> <li>- Allows for the construction of prediction interval without the need for additional model modifications.</li> </ul> | <ul style="list-style-type: none"> <li>- May require multiple forward passes during inference, which can increase computational overhead.</li> <li>- Effectiveness can be influenced by the choice of dropout rate and model architecture.</li> </ul>                     |
| <b>Extreme Learning Machine (ELM)</b><br>[52]                          | Machine Learning | <ul style="list-style-type: none"> <li>- Fast learning speed and scalable.</li> <li>- Simple training process without iterative tuning.</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>- Limited interpretability of model internals.</li> <li>- Performance may suffer in imbalanced data.</li> </ul>                                                                                                                    |

power grids [104]. In solar energy, models like Quantile Regression Forest (QRF) provides a clearer understanding of uncertainty, helping stakeholders optimise energy production management by accounting for a broader range of possible outcomes [85]. Hybrid physics–ML frameworks [105] have shown remarkable success across domains, such as manufacturing force prediction [106], and similar strategies are being adapted for renewable energy.

The use of advanced UQ methods such as PI and are essential for renewable energy forecasting. By addressing both data-driven and model-driven uncertainties, these methods provide a comprehensive understanding of potential energy production outcomes [13], [107]. The application of PI across various renewable energy sources offers more accurate and reliable tools for managing energy resources, ensuring grid stability, and improving operational efficiency. As renewable energy continues to expand, the role of UQ methods like PI will become increasingly indispensable, making them critical for the future of sustainable energy systems [108], [109].

#### IV. APPLICATIONS OF PREDICTION INTERVAL IN RENEWABLE ENERGY FORECASTING

The implementation of optimised aggregation methods enhances forecasting models by generating prediction interval that not only provides point estimates but also quantify uncertainties and disturbances associated with the forecasts [14]. This approach offers crucial insights into the range of possible outcomes and the confidence level in the predictions, significantly improving risk management and decision-making processes. In renewable energy forecasting, where sources like solar, wind, tidal, and hydro energy are subject to inherent variability and uncertainty, such enhancements are particularly valuable [110], [111].

For instance, in renewable energy systems, neural network-based PI is employed to forecast energy production, demand trends, and grid stability. By quantifying uncertainties through PI, energy providers and grid operators are better equipped to optimise resource allocation, manage fluctuations in energy supply and demand, and make informed decisions that improve the efficiency and reliability of renewable energy infrastructure [15]. In power grids, PI generated

by neural network help predict electricity consumption, grid congestion, and potential system failures. This capability enables operators to anticipate risks, optimise grid operations, and ensure the stability and resilience of power distribution networks [74].

Moreover, the application of optimised aggregation methods extends beyond renewable energy. For example, financial analysts can use this approach to generate more robust PI for forecasting asset prices, market trends, and risk assessments [112]. In energy demand prediction, the method improves the accuracy of forecasts related to electricity consumption, renewable energy production, or grid stability. Conformal prediction has also been applied beyond renewable generation, for example in electricity price forecasting for day-ahead and real time balancing markets [113]. These enhancements support efficient resource allocation and strategic energy management, ultimately leading to better economic and operational outcomes in energy systems [14].

In this section, we will describe into the applications of various PI estimation methods across different types of renewable energy sources. By examining the data used, the PI estimation methods applied, and the resulting outcomes for each energy source, we aim to provide a comprehensive analysis of how different PI method addresses the unique challenges associated with solar, wind, tidal, and hydro energy. This analysis will offer deeper insights into the advantages and limitations of these methods, thereby enhancing the understanding of their effectiveness in renewable energy forecasting.

## A. PREDICTION INTERVAL IN SOLAR ENERGY FORECASTING

Accurate solar energy forecasting is critical, particularly in regions with variable sunlight exposure [76]. Prediction interval is instrumental in accounting for uncertainties in solar irradiance, enabling better integration of solar power into energy grids. Traditional forecasting methods provide point estimates but fail to capture the uncertainties inherent in solar power generation. PI addresses this gap by offering a range of possible values, thereby quantifying the uncertainties and disturbances that influence solar energy generation [54]. This information is crucial for decision-making processes in solar energy management, especially for operators involved in energy markets, who must optimise bidding strategies based on forecasted solar power outputs under varying conditions [110].

For instance, Interval Type-2 fuzzy model [110] provides an uncertainty range along with the forecasted values, which allows solar plant operators to make more informed decisions regarding the management of solar power generation under different operational conditions. This model enables better planning and risk management by helping operators develop strategies that take into account the inherent uncertainty in solar irradiance forecasts.

In recent years, advanced methods such as neural network-based PI have gained prominence in solar energy forecasting.

These methods allow energy providers and grid operators to manage fluctuations in energy supply and demand effectively. Neural network can predict electricity consumption, grid congestion, and potential failures, thus helping to anticipate risks, optimise grid operations, and ensure stability and resilience in the energy distribution system. For example, Rana et al. [60] introduced the 2D-interval forecast method, which combines variable selection with ensemble neural networks to provide accurate prediction interval. This approach, known as the Neural Network Ensemble in 2-Dimensions (NNE2D), offers not just point forecasts but a range of values, expressed as upper and lower bounds, which significantly improves forecast accuracy for solar PV power generation. Recent comparative studies demonstrate that conformal prediction can rival or outperform standard ML models in very short term solar power forecasting [114].

Another important contribution in the field comes from the work by Galván et al. [115], who used neural network alongside multi-objective particle swarm optimisation to estimate PI for solar energy forecasting. Their approach balances the trade-off between coverage probability and interval width, providing more reliable forecasts. The use of neural network enables the models to capture complex, non-linear relationships in solar energy data, leading to improved forecasting performance.

Studies like those from Voyant et al. [96] have also made significant strides in global solar irradiation forecasting, using Regression Trees method to generate PI. These methods have practical applications in energy grid management, renewable energy integration, and risk assessment. Regression tree-based methods provide flexibility and scalability, making them suitable for large-scale solar energy systems. They enhance the accuracy and reliability of solar energy forecasts, which are crucial for energy market trading and grid stability.

The implementation of optimised prediction interval methods further supports solar energy management by improving the accuracy of solar power generation forecasts. For instance, Ni et al. [116] developed an optimised PI approach using the Bootstrap method and ELM algorithms. This method enhances the accuracy and reliability of short-term solar PV power forecasting. Additionally, the Fully Online Sequential ELM algorithm (FOS-ELM), which incorporates a forgetting mechanism, continuously updates the model to improve forecast reliability and efficiency. These models are instrumental for grid operators, who need accurate and narrow PI to manage the variability of solar power generation [117].

Alcantara et al. [118] compared quantile estimation methods with direct PI estimation, finding that optimising specific loss functions during PI construction significantly improves forecast accuracy and reliability. This comparison highlights the advantages of direct PI estimation in capturing the variability of solar power generation and integrating it into the energy grid. Such models optimise energy trading strategies and improve grid stability by providing precise and narrow PI, which allow for better energy resource allocation and risk management.

Another promising method for enhancing PI is the Improved Bootstrap method, which clusters solar data into sub-datasets and constructs PI through bootstrapping. This method has shown better performance when compared to traditional approaches. Similarly, the LUBE method has demonstrated success in solar power forecasting, as it outperforms traditional models by generating accurate and reliable PI. The integration of LUBE with other advanced techniques like Support Vector Regression (SVR-2D) and ELM has further improved forecast performance, making it a valuable tool for solar energy systems [63], [78].

Finally, advanced DL models like the Quantile-Driven Deep Neural Network (QDDNN) and hyper-networks are used for direct PI estimation, optimising loss functions to generate lower and upper bounds for PI [118]. These models can estimate multiple prediction intervals with a single model, which improve both the efficiency and accuracy of solar energy forecasts. By addressing multi-objective problems and offering a Pareto front of solutions for optimal coverage-width trade-offs, these methods provide significant improvements in forecast accuracy and reliability.

Table 10 presents a concise comparison of datasets and methodologies used in solar energy forecasting, focusing on prediction interval. By offering a range of forecasted outcomes, PI enhance decision-making for solar energy integration into grids, helping manage the uncertainties in solar irradiance. The table summarises key methods like neural network and evolutionary optimisation, giving stakeholders a clear view of the most effective approaches for balancing supply and demand in solar energy systems.

## B. PREDICTION INTERVAL IN WIND ENERGY FORECASTING

Wind energy forecasting presents unique challenges due to the highly variable nature of wind speeds, which directly affects power generation. Accurately predicting wind power output requires the use of advanced forecasting techniques, with prediction interval playing a crucial role [73]. PI provide a range of possible outcomes, improving the predictability and operational efficiency of wind farms by capturing uncertainties and variability in wind power generation. This section explores the application of PI in wind energy forecasting, highlighting advanced methodologies and their advantages, challenges, and applications [12].

To address the inherent uncertainties in wind energy, several techniques have been employed. Liu et al. [68] propose a framework that integrates fuzzy logic, multi-objective optimisation, and PI construction to enhance wind energy predictions. Fuzzy logic handles the uncertainty in wind power forecasting effectively, while multi-objective optimisation addresses conflicting objectives like prediction accuracy and interval width simultaneously [107]. By incorporating these techniques into deep learning models, the framework optimises multiple goals such as accuracy, generalisation, and efficiency, resulting in more robust and reliable predictions [119]. This method enhances decision-making in grid man-

agement and helps energy providers plan better for fluctuations in power output.

Wind energy forecasting has seen significant advancements with methods such as the LUBE combined with Radial Basis Function Neural Network (RBFNN), LSTM Network [108], Empirical Mode Decomposition (EMD), Sample Entropy Technique (SET), and Variational Mode Decomposition (VMD) [120]. These methods contribute to more accurate and reliable wind power forecasts by addressing different aspects of the forecasting process, from data decomposition to neural network-based PI estimation [15]. The integration of neural network and decomposition techniques helps capture the inherent non-linearities and chaotic behaviour of wind energy, improving forecast precision.

For example, a combination of LUBE and RBFNN has been used to estimate PI with high coverage probability, allowing operators to better anticipate the range of possible wind power outputs. In another study, the LSTM network, which excels in capturing temporal dependencies in sequential data, is combined with EMD and VMD for wind power forecasting. These methods not only enhance forecast accuracy but also provide sharp and reliable PI, aiding in the smooth integration of wind energy into the power grid [108].

The application of multi-objective optimisation algorithms has greatly improved wind energy forecasting models by balancing prediction accuracy and interval width. This approach minimises forecasting errors while enhancing the reliability of wind power predictions. For instance, Wang et al. [103] developed a combined model using VMD and multi-objective optimisation [30] to optimise wind power generation systems. This model aids decision-makers in scheduling and operation, thus improving the overall efficiency of wind energy utilisation.

Heuristic optimisation algorithms, such as the Particle Swarm Optimisation (PSO) and the Modified Bat Algorithm (MBA), have also been used in conjunction with PI construction techniques [119]. These algorithms optimise model parameters, improving the performance of forecasting models. For instance, Wan et al. introduced a Hybrid Intelligent Algorithm (HIA) combining ELM and PSO to directly formulate optimal PI for wind power generation [103]. The integration of these two techniques results in more precise and reliable wind power predictions, which are essential for energy grid management and renewable energy integration.

Additionally, methods like the center-of-concentration based neural network approach help improve the reliability of wind power forecasts by accounting for uncertainties in the data [121]. This method, along with the use of non-parametric models and ensemble techniques, provides accurate PI that enable wind farm operators to manage energy production more effectively [122].

Several advanced models have been developed to further enhance the prediction of wind power generation. For instance, the combination of VMD and Kernel Extreme Learning Machines (KELM) [123], along with an improved artificial bee colony algorithm, has been applied to generate

TABLE 10: Application of Solar Energy in Forecasting Models

| Cite  | Dataset                                                 | Source                                            | Methods                                                                                     | Outcome                                                        | Limitation                                                              | Recommendations                                                                   |
|-------|---------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| [60]  | Australian solar PV power data (2013-2014)              | Australian solar PV records                       | NEE2D method                                                                                | Outperformed baselines in accuracy and coverage probability    | Data pre-processing, feature selection, variability and uncertainty     | Explore sensitivity analysis, normalisation methods, apply to other energy series |
| [115] | Solar energy prediction contest data                    | Kaggle (Online)                                   | Multi-objective evolutionary optimisation, neural networks                                  | Accurate and informative forecasts through prediction interval | Trade-off between coverage probability and interval width, data quality | Enhance model interpretability, improve pre-processing, ensemble modelling        |
| [85]  | Global Energy Forecasting Competition 2014 (GEFCom2014) | Three solar stations in Australia                 | Multi-objective particle swarm optimisation, Neural Network (NN)                            | Reduced uncertainty in short-term forecasts                    | Data quality, model complexity                                          | Improve data quality, simplify models, optimise techniques                        |
| [96]  | Hourly solar GHI measurements                           | Meteorological station data                       | Regression trees                                                                            | Improved probabilistic forecasts                               | Data quality, scalability                                               | Explore advanced machine learning techniques                                      |
| [63]  | Solar data of UQ center                                 | University of Queensland                          | Improved Bootstrap, Envelope-based cluster technique                                        | More accurate and narrower prediction interval                 | Data purity, prediction accuracy, dataset complexity                    | Extend methodology to other renewable sources                                     |
| [78]  | Realistic PV power data                                 | DKA solar center                                  | Combination interval prediction model based on biased convex cost function and auto-encoder | Improved interval prediction performance                       | Data availability, hyper-parameter tuning complexity                    | Enhance data collection, explore advanced techniques                              |
| [78]  | Historical solar power and meteorological data          | Australian PV data                                | Extreme Learning Machine, biased convex cost function                                       | Efficient uncertainty quantification                           | Model complexity, hyper-parameter tuning                                | Investigate advanced techniques, enhance interpretability                         |
| [78]  | Solar power output data (2016-2018)                     | Australian PV data                                | Bi-level optimisation, prediction engines                                                   | Superior prediction performance and robustness                 | Data quality, model interpretability                                    | Validate and benchmark models                                                     |
| [117] | Solar Power Generation Data                             | Rooftop PV systems, weather stations              | FOS-ELM algorithm                                                                           | Improved forecast accuracy and reliability                     | Data quality, integration of diverse sources                            | Use advanced preprocessing, ensemble learning                                     |
| [117] | Weather Variables Data                                  | Meteorological stations                           | Deep RNN models                                                                             | High accuracy for short-term forecasts                         | Data synchronisation, feature engineering                               | Hybrid forecasting models                                                         |
| [117] | Environmental Conditions Data                           | Environmental monitoring stations, satellite data | Neural network models                                                                       | Effective PV system power modelling                            | Data heterogeneity, model generalisation                                | Research transfer learning techniques                                             |
| [117] | Solar Energy Production Data                            | Utility-scale solar farms                         | Support vector machine                                                                      | Improved prediction accuracy                                   | Parameter tuning, handling nonlinear relationships                      | Integrate explainable AI techniques                                               |
| [110] | Solar power data from a 75-kW plant                     | NV Energy Corporation                             | Interval Type-2 TSK modelling                                                               | Better forecasts, represented uncertainty                      | Model complexity, data quality                                          | Extend to other TSK systems, integrate weather forecasting                        |
| [110] | Meteorological data                                     | Local weather stations                            | TSK modelling                                                                               | Qualitative models for power prediction                        | Data quantification, model interpretability                             | Advanced uncertainty modelling, real time data integration                        |
| [110] | Measured data sampled at 1 sample/minute                | NV Energy Corporation                             | Comparison of TSK models                                                                    | Better representation of uncertain data                        | Model calibration, dynamic nature of solar energy                       | Operational implementation, practical validation                                  |
| [118] | Renewable Energy Production Data                        | Meteorological stations, energy plants            | Deep Neural Network (DNN)                                                                   | Improved regional forecasting accuracy                         | Data quality, historical data availability                              | Enhance data quality, explore ensemble techniques                                 |
| [118] | Lugo, Burgos, Córdoba Energy Data                       | Regional energy records                           | DNN with gradient descent optimisation                                                      | Successful regional forecasting                                | System integration, regulatory compliance                               | Enhance interpretability, incorporate external factors                            |
| [20]  | Solar Power Data                                        | Local Weather Stations                            | Deep Learning (DCNN, DRNN, DBN, SAE)                                                        | Improved Forecasting Accuracy                                  | Model Complexity, Data Interpretability                                 | Integration of Multiple Algorithms, Unified Predictive Models                     |

accurate and reliable PI [124]. These models are particularly useful for short-term wind speed prediction, aiding in the scheduling of wind power production and its seamless integration into the power grid.

In addition, ensemble techniques such as the Gated Recurrent Unit (GRU) based ensemble model have been employed for short-term wind speed interval prediction. This method not only enhances wind power production schedules but also supports better decision-making for grid integration and energy management [125].

Moreover, the ANEMOS wind power prediction platform, operational worldwide, uses resampling techniques to im-

prove forecasting accuracy. It incorporates meteorological forecasts from models like High Resolution Limited Area Model (HIRLAM) to enhance the reliability of wind power predictions [126]. These platforms and models underscore the importance of using advanced forecasting techniques to manage the inherent variability in wind energy production.

While significant progress has been made, wind energy forecasting using PI still faces challenges. The high variability and unpredictability of wind speeds pose difficulties in constructing reliable prediction models [12]. The need for accurate data, advanced algorithms, and efficient computational resources are critical factors in improving forecast accuracy.

However, the combination of advanced machine learning models, optimisation techniques, and heuristic algorithms provides a promising direction for future improvements.

Table 11 and 12 highlights the datasets and techniques applied in wind energy forecasting. The role of PI in capturing the unpredictable nature of wind speeds is emphasised, with advanced methods like LUBE and multi-objective optimisation enhancing accuracy. These tables provide snapshots of how different approaches contribute to better grid integration and energy distribution strategies, enabling operators to handle the challenges of wind variability more effectively.

### C. PREDICTION INTERVAL IN TIDAL ENERGY FORECASTING

Tidal energy generation is inherently influenced by complex ocean conditions and tidal patterns, leading to variability in energy output. Prediction interval provides a statistical framework for estimating the range of possible tidal energy outputs, significantly improving the accuracy and reliability of tidal energy forecasts. One of the prominent methods for tidal energy forecasting is the Harmonic Residual Analysis (HRA) model, which integrates techniques from Information and Fractal Theory to enhance the precision of tidal current predictions [129]. This hybrid model improves existing numerical schemes used in tidal energy installations, offering a reliable short term online forecast solution. The HRA model enables better grid integration, making tidal energy a more dependable renewable energy source by reducing forecasting errors and operational risks.

Prediction Interval also plays a vital role in tidal energy resource assessment, helping developers identify suitable sites for tidal energy extraction. A model for predicting tidal velocity in fjord entrances, gateway areas for tidal currents, demonstrates how accurate forecasts of tidal velocities contribute to optimising tidal system design. This model also supports environmental impact assessments, where precise tidal forecasts help evaluate the effects of tidal energy extraction on local ecosystems, sediment transport, and water quality. Such comprehensive forecasting allows developers to balance energy generation needs with environmental sustainability [130].

The Support Vector Regression (SVR) model combined with PI is another promising technique for forecasting tidal current variations. This model improves the reliability of tidal current energy generation forecasts by incorporating uncertainty quantification, enabling better decision-making in tidal energy production and grid management [131]. The integration of SVR and PI enhances system optimisation by maximising the utilisation of tidal resources and improving the sustainability of tidal energy generation.

Multi-objective optimisation algorithms have also been employed to address the uncertainty inherent in tidal energy forecasting [132]. These algorithms are effective in optimising model parameters to achieve accurate and reliable predictions. For instance, the LUBE method combined with SVR has been successfully used to construct prediction interval

for tidal current forecasts [131]. This method, coupled with the Flower Pollination Algorithm, optimises SVR parameters, thereby refining prediction interval and improving the accuracy of tidal current forecasts.

Harmonic Analysis, Singular Spectrum Analysis (SSA), and Linear Recurrent Forecasting (LRF) are additional techniques used to decompose and forecast tidal energy data [133]. These methods contribute to noise reduction, signal decomposition, and reliable prediction interval for tidal currents, further enhancing the accuracy of tidal energy generation systems. High Order Fuzzy Time Series (HOFTS) model has also been implemented to improve short-term forecasts of tidal currents by providing robust prediction interval that account for the variability in ocean conditions.

Evaluating the turbine technology selection process is another critical aspect of tidal energy forecasting. The decision between turbines with yaw systems and fixed-axis turbines depends heavily on the predictability of tidal currents. By using prediction interval, developers can make more informed choices regarding which turbine technology is most suitable for a specific tidal energy site, optimising energy extraction efficiency [80].

In Monahan et al. [129] study, a Non-parametric Prediction Interval (NPI) model combined with Bi-level Optimisation Formula for Extreme Learning Machine (BOF-ELM) was applied to forecast tidal currents and levels. This methodology integrates various forecasting techniques, such as Harmonic Analysis for tidal current prediction and SSA for signal decomposition, to provide high-accuracy short term tidal energy forecasts. These hybrid models offer substantial improvements in forecasting accuracy, which is essential for optimising tidal turbine placement and operation, ultimately enhancing grid integration and stabilising the energy supply.

Table 13 outlines the datasets and methodologies involved in tidal energy forecasting. This table focuses on how hybrid models like HRA and SVR-based optimisation techniques improve forecast reliability. These approaches are essential for optimising turbine placement and resource management in tidal energy systems, ensuring efficient and sustainable use of tidal resources.

### D. PREDICTION INTERVAL IN HYDRO ENERGY FORECASTING

Hydro energy generation, driven by fluctuating water flow rates, presents significant variability, which can challenge the accuracy of energy forecasts. Prediction interval is an essential tool in addressing this variability, providing a range of possible outcomes that enable more precise hydro energy forecasts and support better resource management and operational planning [98]. This section explores various advanced methodologies for hydro energy forecasting that utilise PI to optimise energy production and enhance decision-making in the energy sector.

Zhang et al. [135] introduced a cutting-edge forecasting system that integrates hydropower generation into multi-energy complementary systems using LSTM network. This

TABLE 11: Application of the Wind Energy in Forecasting Models (Part 1)

| Cite  | Dataset                                                 | Source                                                                                                               | Methods                                                                                                                                                            | Outcome                                                                                                 | Limitation                                                                                          | Recommendations                                                                                                                                    |
|-------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| [103] | Wind power generation data                              | Challicum Hills and Starfish Hill wind farms in Australia                                                            | Hybrid Intelligent Algorithm (HIA) approach for optimal prediction interval                                                                                        | Reliable prediction interval with high confidence, outperforming traditional methods                    | Nonstationarity and heteroscedasticity of wind power series                                         | Enhance flexibility and regression ability of HIA approach, explore applications in reserve determination, wind power trading, and unit commitment |
| [126] | Wind power point forecasts and measurements Data        | Danish wind farm located in Klim, North Jutland                                                                      | Statistical forecasting methods including Sipreóico, AWPPS, WPPT using meteorological forecasts from HIRLAM with conditional prediction interval estimation method | Improved accuracy in representing uncertainty                                                           | Infrequent cut-off events impacting probabilistic forecasts                                         | Address impact of other meteorological variables on forecast uncertainty, enhance probabilistic forecasting models                                 |
| [107] | Kinmen wind turbine data                                | Online data                                                                                                          | Center-of-concentrated-based neural network method                                                                                                                 | Improved prediction interval bound, similar coverage probability to traditional methods                 | Assumptions of data distribution, large computational complexity                                    | Optimise prediction interval for multistep-ahead forecasting, explore heuristic optimisation algorithms                                            |
| [124] | Cathedral Rocks farm dataset                            | Wind Farm Australia [127]                                                                                            | Hybrid approach combining EEMD, ELM-DIC, and QPSO                                                                                                                  | Improved quality of prediction interval with higher reliability and sharpness                           | Non-stationary and noisy nature of wind power time series                                           | Enhance construction of optimal prediction interval for wind power generation                                                                      |
| [124] | Woolnorth wind farm dataset                             | Wind farm power generation recorded in 5-minute intervals, Australia [12]                                            | Ensemble empirical mode decomposition and ELM technique                                                                                                            | Enhanced reliability and sharpness of wind power prediction interval                                    | Intermittency and variability of wind power                                                         | Refine prediction interval methods for wind energy forecasting                                                                                     |
| [124] | NREL wind farm dataset                                  | National Renewable Energy Laboratory                                                                                 | Ensemble empirical mode decomposition and sample entropy techniques                                                                                                | Increased effectiveness in wind power interval forecasts                                                | Uncertainty and variability in wind power generation                                                | Advance techniques for constructing optimal prediction interval to improve reliability and accuracy                                                |
| [119] | Woolnorth and Starfish Hill wind farm data in Australia | Wind farms located in Australia                                                                                      | Fuzzy-based combined prediction interval using the LUBE method and modified Bat Algorithm                                                                          | Improved accuracy in wind power forecasting with optimal prediction interval                            | Complexity of forecasting data, nonlinearity of power systems, bias in parametric-based forecasters | Explore hybrid methods combining statistical and intelligent models for enhanced forecasting accuracy                                              |
| [108] | Historical wind power generation data                   | Adelaide and Bornish wind farms in Ontario, Canada (IESO website)                                                    | recurrent neural network (RNN) and long short-term memory (LSTM)                                                                                                   | LSTM outperformed typical RNN models in real-world data sets                                            | Incorporating uncertainty in wind power forecasting, computation-intensive methods                  | Explore deep learning models like LSTM for wind power forecasting, improve uncertainty estimation and prediction interval generation               |
| [12]  | Monthly datasets of the wind farm (2010-2011)           | Wind farm power generation recorded in 5-minute intervals, Australia                                                 | Neural network-based models for short-term wind power forecasting                                                                                                  | Reliable prediction interval for different forecasting horizons                                         | Inherent and irreducible uncertainties in wind power forecasts                                      | Incorporate additional variables like meteorological data and turbine characteristics to improve forecast accuracy and reduce uncertainty          |
| [128] | Wind speed data and Renewable Energy Sources and        | Wind Speed from Sotavento wind farm in Spain and Renewable Energy Sources from North China Electric Power University | Lorenz disturbance distribution for wind speed prediction                                                                                                          | Improved wind speed prediction accuracy and interval estimation                                         | Dependency of interval prediction on accuracy of point prediction models                            | Research universal applicability to multiple wind farms and diverse scenarios to enhance prediction precision                                      |
| [73]  | Wind power generation data of 2010                      | Woolnorth and Capital Wind Farm in Australia                                                                         | Nonparametric method for combined Lower Upper Bound Estimation (LUBE) prediction interval using neural network forecasts                                           | Reliable prediction interval for wind power generation, outperforming traditional parametric approaches | High variability and nonlinearity of wind power data                                                | Improve stability and consistency of neural network models for accurate wind power forecasting                                                     |
| [30]  | Wind speed data of Shandong Province                    | Three wind farms in Shandong Province                                                                                | Variational Mode Decomposition (VMD) combined with Multi-Objective Optimisation Algorithm                                                                          | Improved accuracy in short-term wind speed interval prediction                                          | Non-stationary and chaotic characteristics of wind speed data                                       | Enhance robustness and applicability of the interval prediction model                                                                              |
| [123] | Wind power data                                         | M and J wind farm in China                                                                                           | VMD and Kernel Extreme Learning Machine optimised by Improved Artificial Bee Colony (IABC) algorithm                                                               | Superior wind power interval prediction accuracy                                                        | Optimal number of modes for VMD, over-decomposition or noise issues                                 | Refine VMD decomposition process, optimise parameters for more accurate wind power interval predictions                                            |
| [125] | Offshore wind field data                                | Boston in 2011 and 2012                                                                                              | Ensemble GRU model combining VMD and error correction                                                                                                              | Improved prediction interval quality in terms of PICP and PINAW                                         | Restrictive assumptions about data distribution, computational loads                                | Enhance ensemble models with preprocessing and post-processing techniques                                                                          |
| [125] | Weekly wind speed data                                  | Offshore wind field in Boston                                                                                        | Comparison of ANN, GRUM, GRU-EC, and VMD-GRU-EC models                                                                                                             | VMD-GRU-EC model outperformed others in terms of CWC index                                              | Stability and accuracy in long term wind speed interval prediction                                  | Improve ensemble models for mass data validation and long-term forecasting                                                                         |

TABLE 12: Application of the Wind Energy in Forecasting Models (Part 2)

| Cite  | Dataset                                        | Source                                                                                           | Methods                                                                                                                | Outcome                                                                               | Limitation                                                                                                            | Recommendations                                                                                                                                                     |
|-------|------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [121] | Eastern wind dataset                           | National Renewable Energy Laboratory (NREL) website                                              | Multi-objective framework with SVM-based LUBE method                                                                   | High-quality prediction interval for wind speed forecasts                             | Intermittent nature and unpredictability of wind speed, seasonal variations                                           | Refine multi-objective framework, address challenges in wind speed prediction accuracy and uncertainty quantification                                               |
| [122] | Wind speed and wind power data                 | Online                                                                                           | Hybrid approach combining VMD and Low Rank Multi-Kernel Ridge Regression (MKRR)                                        | Improved prediction interval for wind speed and wind power forecasting                | Optimisation of parameters for individual kernels in MKRR, efficient decomposition with VMD, computational complexity | Optimise hybrid model, explore additional decomposition techniques, integrate advanced optimisation algorithms, validate on larger datasets                         |
| [95]  | Real-world Wind Power Dataset                  | Online                                                                                           | LSTM-based LUBE model, RBF Neural Network, EMD and Sample Entropy Techniques, VMD and Partial Autocorrelation Function | Improved accuracy in wind power interval forecasting                                  | Uncertainty in wind power generation, computational complexity, capturing complex patterns                            | Explore ensemble methods, integrate external factors, develop adaptive forecasting models                                                                           |
| [102] | Historical Wind Power Generation Data          | Online                                                                                           | VMD for signal decomposition, Machine Learning Models for forecasting, QRA for probabilistic interval prediction       | Improved short-term wind power forecasting accuracy                                   | Parameter tuning, Noise sensitivity, Computational complexity                                                         | Explore advanced signal processing, enhance machine learning models, investigate ensemble methods, conduct real-world validation                                    |
| [101] | Beijing Air Quality Multivariate Dataset (AQD) | UCI Repository                                                                                   | Deep learning architectures CNN, LSTM                                                                                  | Improved accuracy and reliability of time series forecasting with prediction interval | Computational complexity, model interpretability, uncertainty estimation methods                                      | Enhance computational efficiency, standardise interpretation, explore ensemble techniques, study decision-making impact                                             |
| [104] | Wind power generation data SA and NSW          | South Australia (SA) and New South Wales (NSW) regions, Australian Energy Market Operator (AEMO) | Extreme Learning Machines (ELM) and Self-Adaptive Evolutionary Learning Machines (SAEELM)                              | Improved quality of prediction interval, enhanced reliability and sharpness measures  | Scalability to larger datasets, implementation considerations                                                         | Enhance scalability and robustness, comparative analyses with state-of-the-art methods, address limitations, explore applications in other renewable energy domains |

method enhances the understanding of the variability and uncertainty in renewable energy systems, including wind, solar, and hydropower, empowering operators and planners with tools to make better-informed decisions for scheduling and resource allocation [135]. By employing LSTM network, this approach provides superior accuracy for forecasting hydro energy output, offering a deeper understanding of the range of possible outcomes.

Dehghani et al. [130] revolutionised the field with the Grey Wolf Optimisation-Adaptive Neuro-Fuzzy Inference System (GWO-ANFIS) model, which is applied for real time hydropower generation forecasting. This model enhances operational efficiency by integrating water resource data, such as precipitation, inflow, and historical generation data, to improve the accuracy of predictions. The combination of GWO with ANFIS allows for real time adjustments to hydropower systems, helping to optimise energy production and resource management [130]. Similarly, Chen et al. proposed a multi-time-scale power prediction model for hydropower stations, which accounts for uncertainties in reservoir inflows, electricity prices, and consumption rates. Their model combines Multi-time scale Coupling Operation (MCO), Dynamic Bayesian Network (DBN), and Probability-based Prediction (PBP), resulting in a robust framework for hydro energy forecasting [136].

In another study, Zhu et al. [137] explored the use of confidence interval models for renewable energy optimisation and

risk assessment in power system scheduling. These models offer valuable insights into the uncertainty of hydro energy production, allowing operators to plan and allocate resources more efficiently. Additionally, Barzola et al. [138] applied ANN to forecast hydroelectric production in Ecuador. By optimising hyper-parameters and using data transformation techniques, they achieved highly accurate forecasting results, which are crucial for effective energy management.

The significance of uncertainty quantification is further emphasised by Nourani et al. [139], who presented methods for improving prediction interval in hydro-climatic modelling. Their study evaluated various PI estimation approaches, including Bayesian, Monte Carlo, Bootstrap, and LUBE methods. This comprehensive evaluation provides insights into the relative strengths and weaknesses of each technique when applied to hydrological and climate-based energy forecasting [139]. These methods allow operators to quantify the uncertainty associated with water flow rates and other environmental variables, thereby improving the reliability of hydro energy forecasts.

A notable application of PI is presented by Liu et al. [140], who integrated wind and hydro energy systems into a real time energy management system. By optimising the coordination between wind and hydro units, their model ensures better grid stability and enhances the utilisation of renewable energy. Meanwhile, Tang et al. introduced the Global Forecast System Quantitative Precipitation Forecast (GFS-QPF)

TABLE 13: Application of the Tidal Energy in Forecasting Models

| Cite  | Dataset                                                           | Source                                                                   | Methods                                                                                      | Outcome                                                                                                                                                                                                           | Limitation                                                                                                                                                                     | Recommendations                                                                                                                                                                                          |
|-------|-------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [129] | Tidal current data from Acoustic Doppler Current Profilers (ADCP) | Real time measurements sites in the United Kingdom and the United States | Harmonic Residual Analysis (HRA) model incorporating Information and Fractal Theory          | Improved accuracy in short-term tidal current predictions                                                                                                                                                         | Data leakage, model overfitting, computational intensity of deep learning methods                                                                                              | Integration of probabilistic machine learning methods, optimisation of training timescales, exploration of ensemble models                                                                               |
| [134] | Turbulence datasets from High frequency AD2CP measurements        | Banks Strait and Clarence Strait in Australia                            | Estimation of turbulence parameters from beam velocities using ADCPs                         | Enhanced understanding of turbulence intensities, TKE densities, and integral length scales at the sites                                                                                                          | Lack of specifications for turbulence measurements and parameter estimation                                                                                                    | Development of new guidelines for turbulence characterisation in tidal energy candidate sites, leading to more reliable results for turbine performance models and device long term integrity            |
| [132] | ADCP measurements in the tidal channel Skarpsundet                | Data collected during field measurements in Norway                       | Comparison of model-predicted tidal velocities with actual ADCP measurements                 | Discrepancies between model predictions and ADCP measurements, highlighting the need for further refinement of the model                                                                                          | Variability in tidal flow patterns, potential measurement errors, and the need for more accurate data for model validation                                                     | Enhance the model by incorporating more detailed tidal flow patterns and local topographic features to improve accuracy in predicting tidal velocities in fjord entrances                                |
| [131] | Tidal current dataset Bay of Fundy                                | Practical data Bay of Fundy, NS, Canada                                  | Support Vector Regression (SVR) combined with the Lower Upper Bound Estimation (LUBE) method | Effectively constructs optimal prediction interval for tidal current speed and direction, improving both Prediction Interval Coverage Probability (PICP) and Prediction Interval Normalised Average Width (PINAW) | High nonlinearity and complexity of the SVR structure, initialisation of the SVR model, and optimisation algorithm can impact the consistency and stability of the LUBE method | Investigating further enhancements in combining SVR models for improved accuracy and exploring advanced optimisation algorithms for better stability and reliability in constructing prediction interval |
| [133] | Tidal current data Philadelphia (2007-2009)                       | Philadelphia station, USA                                                | Harmonic Analysis of Least Square (HAPLS) and model-free estimators                          | Improved prediction accuracy of tidal current speeds several years into the future                                                                                                                                | Error accumulation in myopic prediction methods, sensor misreads, and Eddie currents affecting data quality                                                                    | Integration of conventional tide forecasting techniques with model-free estimators for more consistent and accurate predictions                                                                          |
| [80]  | Power production of Fromveur Strait                               | Fromveur Strait, France                                                  | Three-dimensional (3D) numerical approach(TELEMAC) and Linear approximation method           | Comparison of power production predictions for yawed and fixed-axis turbines using different computational methods for predicting tidal current production                                                        | Overestimation of current velocities and power during extreme tidal cycles with the linear interpolation method                                                                | Refining computational methods to improve accuracy during extreme tidal conditions and enhance overall prediction capabilities                                                                           |

for hydropower generation scheduling. Their method uses precipitation forecasts to improve reservoir operations and optimise water system management, highlighting the value of PI in enhancing hydropower forecasting [141].

Table 14 provides a comprehensive summary of the datasets, methods, and outcomes associated with different hydro energy forecasting approaches. With techniques like LSTM network and GWO-ANFIS models, the table showcases how prediction interval is leveraged to improve hydropower generation forecasts, aiding in better resource management and grid stability.

## E. PREDICTION INTERVAL IN RENEWABLE ENERGY GRID

Energy grids, tasked with balancing supply and demand, face significant challenges due to the variability in input from renewable energy sources. PI offers a robust solution for handling this variability, allowing grid operators to make informed decisions and enhancing the overall stability and efficiency of the grid. Through interval forecasting, grids can account for the uncertainties associated with energy sources like wind, solar, and hydropower. The use of decision trees and interval forecasting methods, such as Gaussian Prediction Interval and Quantile Interval, has proven effective in modelling the ratio of renewable energy to coal supply, offering

precise interval coverage and narrow interval width. This not only enhances the prediction accuracy but also fosters strategic resource allocation and energy planning for a grid that relies increasingly on renewable sources [77] [142].

Incorporating ML and DL techniques into renewable energy forecasting opens up a wide array of applications with notable implications for energy management [143]. These advanced models improve grid stability by optimising the integration of renewable energy sources and supporting decision-making processes for energy trading, storage optimisation, and resource planning [21]. By providing more accurate forecasts, PI empowers grid operators to better manage energy flows and meet demand with increased confidence. Furthermore, PI assists in demand response programs, incentivising consumers to modify electricity consumption patterns based on predicted availability of renewable energy, thus contributing to a more efficient and sustainable energy system [144].

A key application of prediction interval is in real time energy management systems. These systems rely on accurate forecasting to ensure stable grid operations, especially when integrating fluctuating renewable energy sources. For instance, by utilising deep learning based prediction models, energy operators can forecast wind and solar energy production as well as load demand, thereby enhancing the efficiency

TABLE 14: Application of the Hydro Energy in Forecasting Models

| Cite  | Dataset                                                                                                              | Source                                                                    | Methods                                                                                                                                                                               | Outcome                                                                                                                                                                                                                                        | Limitation                                                                                                                                                                                                               | Recommendations                                                                                                                                                                                                                 |
|-------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [130] | Hydro power generation dataset                                                                                       | Iran's Water Resources Management Company and Meteorological Organisation | GWO coupled with ANFIS                                                                                                                                                                | GWO-ANFIS outperformed ANFIS in all 20 combinations, showing superior forecasting capabilities for hydropower generation                                                                                                                       | Uncertainty in forecasting precipitation and inflow to the dam, as well as the need for optimisation algorithms to enhance predictive models                                                                             | Explore the potential of ensemble machine learning models and hybrid prediction models, optimise model complexity, and incorporate additional factors like climate change and drought variations for more accurate predictions. |
| [136] | Reservoir historical power generation data                                                                           | CTankeng hydropower station in the Oujiang river basin, China             | Development of a multi-time-scale coupling model with a dynamic Bayesian network (DBN)                                                                                                | The model provided probability-based power predictions with corresponding occurrence probabilities, enhancing decision-making for hydropower station operations                                                                                | Addressing uncertainties in power demand, electricity prices, and reservoir inflows to improve the accuracy of power predictions                                                                                         | Exploring additional factors influencing power generation, and enhancing the model's capability for bidirectional inference                                                                                                     |
| [137] | Historical measured data                                                                                             | region in western Sichuan province                                        | Confidence interval model and Non-parametric kernel density estimation method                                                                                                         | Probability distribution curves were fitted using the non-parametric kernel density estimation method, deriving confidence interval at different confidence levels for renewable energy output and load demand                                 | The fluctuation and randomness of renewable energy output and load demand pose challenges to the safe and reliable operation of power systems                                                                            | Characterising the uncertainty of renewable energy output and load demand through probabilistic forecasts.                                                                                                                      |
| [138] | Monthly Generation Production (MGP) of hydroelectric systems (GWh) and total average monthly precipitation (mm) data | Historical data from Ecuador                                              | Application of artificial neural network (ANN) models, Multilayer Perceptron (MLP) and Long Short-Term Memory (LSTM) network, for time series forecasting of hydroelectric production | MLP models outperformed LSTM models in terms of accuracy for one-step scenarios, with differentiation techniques enhancing prediction accuracy. LSTM models showed improved accuracy in multi-step scenarios with multiple regressor variables | Computational time required for LSTM models was higher compared to MLP models. Fine-tuning hyper parameters and selecting the optimal neural network structure were crucial challenges in achieving accurate predictions | Comprehensive comparison of different deep learning models, such as LSTM and seq2seq, with the Deep Hydro framework. Impact of varying hydrological conditions could enhance the forecasting capabilities of the models.        |
| [140] | Wind speed and water inflow data from February 2011                                                                  | Yunnan, China                                                             | Auto regressive moving average (ARMA) models                                                                                                                                          | Improved coordination between wind farms and hydro units, leading to optimised bidding strategy and reduced profit uncertainty                                                                                                                 | Limited deregulation in the Chinese electricity market, potential constraints in volume limits for hydro units                                                                                                           | Incorporating more renewable energy sources, enhancing interval optimisation models for larger-scale energy systems                                                                                                             |
| [141] | Global Forecast System Quantitative Precipitation Forecasts (GFS-QPFs)                                               | American National Oceanic and Atmospheric Administration (NOAA)           | Hybrid SDP model (HSDP) and Piecewise SDP framework (PSDP)                                                                                                                            | Improved accuracy in medium-range reservoir inflow predictions and hydropower production                                                                                                                                                       | Incorporating forecasts with various lead-times as hydro logic state variables, dealing with uncertainties in inflow forecasts                                                                                           | Exploration of advanced forecasting techniques for enhanced hydropower optimisation                                                                                                                                             |
| [98]  | Artificial and Real Hydro logic Datasets                                                                             | Online                                                                    | Machine learning techniques, linear regression, locally weighted regression, artificial neural network, and model trees                                                               | Superiority of the new method in estimating prediction interval compared to existing techniques                                                                                                                                                | Data quality, potential overfitting, interpretability of results, scalability, and generalizability                                                                                                                      | Enhancing uncertainty estimation, improving model interpretability, exploring ensemble techniques, and validating across renewable energy forecasting.                                                                          |

of Energy Management System (EMS), particularly in the residential sector [25]. Such improvements are critical for maintaining grid stability and maximising the utilisation of renewable resources.

Ensemble learning methods, particularly those that incorporate neural network, have also been applied to prediction interval generation, significantly boosting the reliability of energy forecasts [145]. By combining multiple models, ensemble learning reduces the error margins associated with single-model predictions, providing grid operators and energy traders with more accurate data for decision-making. These advanced prediction interval methods are essential for integrating renewable energy sources into the grid, optimising energy trading, and planning future resource needs [145].

In a study by Zuniga et al. [99], prediction interval was utilised to forecast electricity demand in the real time elec-

tricity market in Mexico, using a dataset that included 15-minute load demand measurements from a representative load zone. The forecasting model was designed to predict electricity demand two hours ahead, adjusting intervals using association rules and ANN. By refining these intervals, the study successfully enhanced the reliability of load demand forecasts, aiding in more effective grid management and energy market decisions [99].

Table 15 provides a detailed summary of the datasets, sources, methods, outcomes, and limitations associated with renewable energy grid forecasting. This summary serves as a concise reference for understanding the strengths and weaknesses of various prediction interval methods used in energy forecasting.

The integration of prediction interval estimation techniques within renewable energy grids has significant implications

for uncertainty quantification, providing operators and stakeholders with the tools needed to make better decisions in resource allocation and energy distribution [146]. As energy markets increasingly prioritise renewable sources, the need for accurate and reliable forecasting methods becomes even more critical [75].

With the growing adoption of machine learning and deep learning methods, prediction interval estimation method has shown great potential in improving real time energy management, grid stability, and energy trading efficiency. These methods not only enhance forecast accuracy but also empower operators to optimise energy systems in line with renewable energy availability. Moving forward, research in prediction interval modelling is poised to continue driving innovation, particularly in the areas of grid integration, energy storage optimisation, and demand response programs, further solidifying the role of PI in the transition to a sustainable energy future.

### **F. SUMMARY OF PREDICTION INTERVAL METHODS USED IN RENEWABLE ENERGY FORECASTING**

Prediction Interval (PI) estimation methods has gained significant traction in renewable energy forecasting, providing robust means for addressing uncertainty in energy production across various renewable energy sources. The integration of these methods into renewable energy forecasting enhances the accuracy and reliability of predictions, enabling more informed decision-making processes in energy grid management, resource allocation, and sustainable energy development. Table 16 provides a comprehensive summary of various prediction interval methods, the corresponding studies, datasets used, and the type of renewable energy applications explored. The methods presented in the table presents an overview of methods used in renewable energy forecasting, cover a wide spectrum, from neural network ensembles and decision tree models to more sophisticated approaches such as variational mode decomposition and hybrid intelligent algorithms. The variety of approaches highlights the interdisciplinary nature of prediction interval research, spanning solar, wind, tidal, hydropower, and grid energy applications.

Renewable energy forecasting methods are essential for improving prediction accuracy and reliability across various sources like solar, wind, tidal, and hydropower. One notable approach, the Neural Network Ensemble in 2D (NNE2D) method, is effective in enhancing solar energy forecasting, particularly in regions with variable irradiance, while multi-objective evolutionary optimisation and neural networks are widely used to generate optimised PI for solar power, ensuring balanced coverage probability and interval width. Regression and Decision trees have also been successfully applied to solar and grid energy datasets, providing practical tools for managing renewable energy integration into power grids.

Advanced techniques such as ELM, ANN, CNN, LSTM, and hybrid intelligent algorithms are used for both short-term and long-term forecasting of solar, wind, and hydropower energy. These methods efficiently handle large datasets, cap-

ture non-linear relationships, and improve prediction interval accuracy [148]. Moreover, tools like VMD and harmonic analysis, especially in wind and tidal energy, offer accurate decompositions of complex signals, enhancing the reliability of renewable energy production forecasts. Overall, these methods contribute to better grid management, risk assessment, and the effective integration of renewable energy sources into the energy system [149].

By incorporating such advanced models, the prediction of renewable energy generation becomes more precise, offering a robust framework for improving energy system planning and operations. The improved forecasting accuracy supports decision-making, risk management, and operational efficiency, essential for the transition to a more sustainable energy landscape.

## **V. CHALLENGES AND FUTURE DIRECTION**

### **A. CHALLENGES IN THE RENEWABLE ENERGY FORECASTING**

Renewable energy systems, characterised by their variability and nonlinearity, make accurate forecasting particularly challenging. The prediction interval estimation method, designed to account for the uncertainties in renewable energy generation, plays a crucial role in addressing these complexities. However, applying PI in renewable energy forecasting is fraught with challenges, ranging from data quality issues to model interpretability and real time application constraints.

A fundamental challenge lies in the availability and quality of data. PI relies heavily on reliable datasets to accurately represent the uncertainties in renewable energy sources like solar irradiance, wind speed, and water flow rates. Unfortunately, inconsistencies in data collection methods, incomplete datasets, and discrepancies in measurement techniques can introduce significant biases into forecasting models, ultimately affecting the accuracy of PI. In the renewable energy sector, where environmental conditions shift unpredictably, ensuring access to real time, high-quality data is essential. The absence of standardised measurement practices across the industry further complicates the ability to produce robust PI, as diverse datasets lead to unreliable forecasts that hinder energy management and planning.

In addition to data challenges, the computational complexity involved in generating prediction interval is significant. While advanced ML and DL models can enhance PI accuracy by capturing complex, nonlinear behaviours in renewable energy systems, they require extensive computational resources, making real time applications difficult. These models, particularly in scenarios where high-frequency data is needed, demand optimisation techniques that can balance the trade-off between model precision and computational efficiency. For energy systems operating in real time, this remains a major hurdle, as forecasting errors or delays in PI generation can lead to inefficiencies in energy distribution and grid stability.

The interpretability of prediction interval models presents yet another challenge. Many stakeholders have limited insight into how predictions are made and how uncertainties are

TABLE 15: Application of the Energy Grid in Forecasting Models

| Cite  | Dataset                                 | Source                                | Methods                                                                                         | Outcome                                                     | Limitation                                           | Recommendations                                                         |
|-------|-----------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------|
| [147] | UK Grid Energy Supply and Demand Data   | Elexon Portal (Online)                | Decision Trees, Gaussian and Quantile Prediction Interval                                       | High interval coverage accuracy and narrow width            | Complex energy dynamics, uncertain forecasts         | Expand decision tree applications, explore new models, enhance accuracy |
| [144] | Renewable Energy Generation Data        | Public databases (Online)             | Combination of ML/DL algorithms                                                                 | Improved accuracy, better resource planning                 | Data variability, model complexity                   | Develop hybrid models, enhance data quality, adapt in real time         |
| [143] | Renewable Energy Generation Data        | NREL database                         | Time series analysis, ML algorithms                                                             | Improved accuracy with prediction interval                  | Data quality issues, model complexity                | Integrate ensemble methods, real time data, collaborate for standards   |
| [52]  | Ontario Electricity Market, PJM         | Canada                                | ANN, Extreme Learning Machines, Wavelet Transforms                                              | Improved point forecasts and interval                       | Uncertainty in price forecasting, model robustness   | Employ ensemble methods, enhance features, integrate real time data     |
| [97]  | Modified Chen Series and Real Load Data | Artificial and real datasets (online) | Selective Joint Supervision Method, Traditional Joint Supervision Method, Quality-Driven Method | Improved predictive capacities and interval informativeness | Computational costs, system-specific characteristics | Enhance computational efficiency, explore diverse datasets              |
| [145] | Renewable Energy Generation Data        | Public datasets (Online)              | Ensemble learning with neural network                                                           | Improved accuracy and reliability of prediction interval    | Computational complexity, model tuning               | Optimise ensemble, explore architectures, validate                      |
| [99]  | Representative Load Zone Data           | Confidential                          | ANN for forecasts, Association-rule mining for interval                                         | Improved accuracy in load forecasting                       | Model complexity, data quality, resources            | Scale to larger datasets, integrate real time data                      |

quantified. This lack of transparency undermines the trust and utility of PI in decision-making, particularly for policymakers and grid operators who rely on these models to manage resources effectively.

Each renewable energy source also presents unique forecasting challenges when applying PI. Solar energy forecasting, for instance, must contend with rapid fluctuations in irradiance caused by factors like cloud cover, which makes generating narrow and accurate PI difficult. Techniques such as the NNE2D method and ELM have improved PI accuracy, but the highly variable nature of solar energy still limits the reliability of forecasts in high-variability regions. Similarly, wind energy forecasting faces challenges related to the unpredictability of wind speeds, especially in offshore and complex terrain environments. Despite advancements in methods like the Lower Upper Bound Estimation (LUBE) framework, the integration of wind power into energy grids remains a challenge due to the wide prediction interval required to accommodate wind variability.

Tidal energy forecasting, though benefiting from relatively predictable tidal patterns, is complicated by the nonlinearity of ocean currents and environmental factors such as seasonal shifts. Prediction Interval for tidal energy generation must account for a wide range of variables, including water depth and tidal cycles, which require high-resolution data that is often unavailable. Although models such as harmonic analysis and Adaptive Neuro-Fuzzy Inference System (ANFIS) have been applied to improve PI accuracy, the lack of accessible and detailed data continues to impede the development of reliable tidal energy forecasts.

Hydro power forecasting similarly grapples with challenges related to environmental variability. Climate change is increasingly affecting water availability and river flow patterns, which introduces uncertainty into hydropower gener-

ation forecasts. Models like Grey Wolf Optimisation (GWO) combined with ANFIS and DBN have improved the accuracy of PI for hydropower, yet long-term predictions remain highly uncertain due to evolving climate conditions [155]. As technology evolves, forecasting methods must adapt to account for uncertainties in infrastructure development, and production efficiency, making it crucial to develop more flexible PI models for this emerging sector [156] [157].

## B. UNRESOLVED ISSUES:

Despite advancements, several key challenges remain under-explored.

- **Data Integration and Quality:** Renewable energy forecasting continues to suffer from inconsistent or incomplete datasets, leading to unreliable PI estimation. The need for standardised data collection and measurement practices across the industry is critical for improving forecast accuracy.
- **Model Interpretability:** As ML and DL models grow more complex, the challenge of understanding how predictions are made, often referred to as the "black-box" problem, becomes more pronounced. Developing explainable AI (XAI) methods is essential for improving the transparency of PI estimation models.
- **Long-Term Forecasting:** The cumulative nature of forecasting errors over extended periods, particularly in wind and tidal energy forecasting, presents significant challenges. Further research is needed to develop models that can provide reliable long term forecasts while accounting for evolving environmental conditions and climate variability.

Addressing these challenges requires continued innovation in PI estimation methodologies, from improving data governance to enhancing model transparency and computational

TABLE 16: Summary of prediction interval estimation methods used in renewable energy forecasting,types of renewable energy sources application

| Methods                                                   | References      | Datasets Used                                                                                                                                                                                                                                                                               | Type of Energy Work |
|-----------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| NNE2D method                                              | [60]            | Australian solar PV records                                                                                                                                                                                                                                                                 | Solar               |
| Multi-objective evolutionary optimisation, Neural Network | [78] [85] [115] | Solar energy prediction contest data (Kaggle) , GEFCom2014 Task 15 - Three solar stations, Solar power output data (2016-2018) (Australian PV data)                                                                                                                                         | Solar               |
| Regression Trees                                          | [96]            | Hourly solar GHI measurements (Meteorological station),                                                                                                                                                                                                                                     | Solar               |
| Decision Trees                                            | [147]           | UK Grid Energy Supply and Demand Data (Elexon Portal)                                                                                                                                                                                                                                       | Grid Energy         |
| Improved Bootstrap                                        | [63]            | Solar data of UQ center (University of Queensland),                                                                                                                                                                                                                                         | Solar               |
|                                                           | [150]           | Real-world data set collected from Refinery                                                                                                                                                                                                                                                 | Energy Grid         |
| Combination interval prediction model                     | [78]            | Realistic PV power data (DKA solar center)                                                                                                                                                                                                                                                  | Solar               |
| Extreme Learning Machine (ELM)                            | [52] [78]       | Historical solar power and meteorological data (Australian PV data),Solar Power Generation Data (Rooftop PV systems, weather stations)                                                                                                                                                      | Solar               |
|                                                           | [104] [117]     | Wind power generation data SA and NSW (South Australia and New South Wales regions, AEMO)                                                                                                                                                                                                   | Wind                |
|                                                           |                 | Ontario Electricity Market, PJM (Canada)                                                                                                                                                                                                                                                    | Grid Energy Demand  |
| Neural Network models                                     | [12] [107]      | Environmental Conditions Data (Environmental monitoring stations, satellite data),                                                                                                                                                                                                          | Solar               |
|                                                           | [117] [143]     | Kinmen wind turbine data (Online data),Monthly datasets of the wind farm (Australia)                                                                                                                                                                                                        | Wind                |
|                                                           | [144]           | Renewable Energy Generation Data (Public databases), Renewable Energy Generation Data (NREL database)                                                                                                                                                                                       | Grid Energy         |
| Ensemble learning with Neural Network                     | [145]           | Renewable Energy Generation Data (Public datasets)                                                                                                                                                                                                                                          | Renewable           |
| Support vector machine                                    | [117]           | Solar Energy Production Data (Utility-scale solar farms)                                                                                                                                                                                                                                    | Solar               |
| Interval Type-2 TSK modelling                             | [110]           | Solar power data from a 75-kW plant (NV Energy Corporation), Meteorological data (Local weather stations), Measured data sampled at 1 sample/minute (NV Energy Corporation)                                                                                                                 | Solar               |
| Deep Neural Network (DNN)                                 | [117] [20]      | Renewable Energy Production Data (Meteorological stations, energy plants), Lugo, Burgos, Córdoba Energy Data (Regional energy records),Solar Power Data (Local Weather Stations),Weather Variables Data (Meteorological stations)                                                           | Solar               |
|                                                           | [118]           |                                                                                                                                                                                                                                                                                             |                     |
| Hybrid Intelligent Algorithm (HIA) approach               | [103]           | Challicum Hills and Starfish Hill wind farms in Australia                                                                                                                                                                                                                                   | Wind                |
| Statistical forecasting methods                           | [126]           | Danish wind farm located in Klim, North Jutland                                                                                                                                                                                                                                             | Wind                |
| Hybrid approach (EEMD, ELM-DIC, and QPSO)                 | [124]           | Cathedral Rocks wind farm dataset (Wind Farm Australia)                                                                                                                                                                                                                                     | Wind                |
| Ensemble empirical mode                                   | [124]           | Woolnorth wind farm dataset (Australia), NREL wind farm dataset (National Renewable Energy Laboratory)                                                                                                                                                                                      | Wind                |
| Prediction Interval using the LUBE method                 | [73] [95]       | Woolnorth and Starfish Hill wind farm data in Australia, Wind power generation data of 2010 (Woolnorth and Capital Wind Farm, Australia),Eastern wind dataset (National Renewable Energy Laboratory website),Wind speed and wind power data (Online),Real-world Wind Power Dataset (Online) | Wind                |
|                                                           | [119] [121]     |                                                                                                                                                                                                                                                                                             | Tidal               |
|                                                           | [131]           |                                                                                                                                                                                                                                                                                             |                     |
| ANN, CNN, RNN and LSTM                                    | [98] [99]       | Historical wind power generation data (Adelaide and Bornish wind farms, Ontario, Canada),Beijing Air Quality Multivariate Dataset (AQD) (UCI Repository),                                                                                                                                   | Wind                |
|                                                           | [101] [108]     | Monthly Generation Production (MGP) of hydroelectric systems data (Ecuador),Artificial and Real Hydrologic Datasets,                                                                                                                                                                        | Hydropower          |
|                                                           | [138] [151]     | Tabriz Wastewater Treatment Plant Effluent Data (Iran), Representative Load Zone Data (Confidential)                                                                                                                                                                                        | Grid Energy         |
| Variational Mode Decomposition (VMD)                      | [30] [102]      | Wind speed data of Shandong Province (Three wind farms in Shandong Province),Wind power data (M and J wind farm in China),Offshore wind field data (Boston, USA), Weekly wind speed data (Offshore wind field in Boston),Historical Wind Power Generation Data (Online)                     | Wind                |
|                                                           | [122] [123]     |                                                                                                                                                                                                                                                                                             |                     |
|                                                           | [125]           |                                                                                                                                                                                                                                                                                             |                     |
| ELM and SAEELM                                            | [104] [152]     | Wind power generation data SA and NSW (South Australia and New South Wales regions, AEMO)                                                                                                                                                                                                   | Wind                |
|                                                           | [153]           | Ontario Electricity Market, PJM (Canada)                                                                                                                                                                                                                                                    |                     |
| Harmonic Analysis model (HRA,HAMLS)                       | [129] [133]     | Tidal current data from Acoustic Doppler Current Profilers (ADCP),Tidal current data Philadelphia (Philadelphia station, USA)                                                                                                                                                               | Tidal               |
| GWO coupled with ANFIS                                    | [130]           | Hydropower generation dataset (Iran)                                                                                                                                                                                                                                                        | Hydropower          |
| Dynamic Bayesian Network (DBN)                            | [136]           | Reservoir historical power generation data (Oujiang river basin, China)                                                                                                                                                                                                                     | Hydropower          |
| Confidence Interval model (CI)                            | [137]           | Historical measured data (western Sichuan province)                                                                                                                                                                                                                                         | Hydropower          |
| Autoregressive moving average (ARMA) models               | [140]           | Wind speed and water inflow data (Yunnan, China)                                                                                                                                                                                                                                            | Hydropower          |
| Hybrid SDP model (HSDP)                                   | [141]           | Global Forecast System Quantitative Precipitation Forecasts (GFS-QPFs)                                                                                                                                                                                                                      | Hydropower          |
| Improved Teaching Learning Based Optimisation algorithm   | [154]           | Huatacondo microgrid data (Chile)                                                                                                                                                                                                                                                           | Microgrid           |

efficiency. By overcoming these hurdles, renewable energy systems can benefit from more accurate, reliable, and actionable forecasts, which are essential for transitioning to sustainable, low-carbon energy solutions.

### C. FUTURE DIRECTION FOR THE RENEWABLE ENERGY FORECASTING

The journey forward in renewable energy forecasting is integral to achieving global Net-Zero emissions. This transition demands not only the adoption of clean energy but also the refinement of forecasting methodologies to ensure the seamless integration of renewable sources into the grid [158]. Improving prediction interval estimation techniques is vital for handling the inherent uncertainties in renewable energy production. Future research should prioritise refining these intervals through advanced optimisation and sensitivity analyses to enhance grid stability and maintain supply-demand balance.

Expanding these forecasting models to underdeveloped regions in Africa and Asia, with vast renewable potential, presents valuable opportunities [159]. By doing so, researchers can uncover insights into energetic complementarity that optimise regional energy infrastructure. Further, deep learning and hybrid models are essential for capturing the complex, nonlinear behaviours of renewable systems, improving both short term and long term forecasts. Integrating PI methods with storage solutions offers another pathway to bolster grid stability.

Emerging technologies like explainable AI are set to revolutionise forecasting by making complex models more interpretable, fostering trust among energy operators and policymakers [160]. Data quality, governance, and model transparency will remain central themes, aligning forecasting advancements with global sustainability goals.

## VI. CONCLUSION

This study underscores the potential of prediction interval estimation techniques to enhance renewable energy forecasting, particularly by addressing the uncertainty and variability intrinsic to renewable sources like solar, wind, tidal, and hydropower. By embedding PI methods in forecasting models, we enable more accurate and reliable predictions, which is critical for decision-making in energy management as the world moves toward Net-Zero emissions.

The findings emphasise that quantifying uncertainties in renewable energy forecasting is essential to managing the variability of these energy sources. Through examining various PI approaches, from statistical to deep learning models, this research illustrates the adaptability of PI in managing diverse renewable sources, supporting the broader agenda of global sustainability. Key challenges, data quality, computational requirements, and model transparency, highlight areas needing ongoing focus to enhance the robustness of renewable forecasting.

Ultimately, this research contributes a practical framework for improving the reliability of renewable energy forecasts,

advancing PI estimation techniques, and broadening their application across emerging energy domains. By addressing these challenges, the energy sector can make significant strides toward building efficient, sustainable energy systems that are resilient and responsive to future demands. Such advancements not only support a cleaner energy landscape but also fortify the path toward a sustainable, Net-Zero future for society and the planet.

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